Locally Irregular Two-Edge-Colourings of Doubled Graphs: Constructions, Extensions, and Obstructions

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Abstract

We study the conjecture that replacing every edge of a connected simple graph other than K_2 by two parallel copies produces a multigraph whose edges admit a locally irregular two-colouring. We present a computer-assisted proof for all subcubic underlying graphs and develop constructions for several families of unbounded maximum degree. These include sparse direct branch cores, degree-homogeneous three-colourable cores joined by paths, balanced partial subdivisions, parity-separated degree patterns, and clique block trees with explicit interface capacity. Two general sufficient conditions are $\delta(G) \geq 8\chi(G)(2\chi(G)+1) - 7$ and $8\delta(G) \geq 7|V(G)| + 6$; the latter admits red degrees distinct at all vertices and blue degrees distinct at all vertices. We extend the dense construction to prescribed external contributions, forbidden final degrees, and sequential absorption of multiple regions. Complement-path lifts give further infinite families. Exact fixed-target factor criteria, exchange and polynomial identities, and explicit obstructions clarify the limits of several stronger strategies. Complete finite interface certificates and executable checkers accompany the proofs. Classical factor, matching and interpolation tools are explicitly attributed. We also prove a signed-gap lemma with at most two occurrences per variable, use it to colour degree-three/four path–cycle partitions, and derive exact fixed-interface criteria and arbitrary-degree root-port constructions for odd components. The conjecture for arbitrary connected graphs is not proved.

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1 Problem, scope, and provenance

For a finite simple graph G, let ${}^2G$ denote the multigraph in which every original edge has two parallel copies. A colour class is *locally irregular* if the degrees of the endpoints of each of its edges are different in that colour class. Parallel copies may have different colours. Isolated vertices cause no difficulty. The conjecture of Grzelec and Woźniak [1] asks whether

$$G \text{ connected}, \quad G \ncong K_2 \quad \Longrightarrow \quad \operatorname{lir}({}^2G) \leq 2.$$

All statements below concern finite graphs. Original graphs are simple; auxiliary connection graphs may have labelled parallel edges and loops. A loop in a connection graph represents a return path in the original simple graph, rather than an original loop.

The basic language is an edge weight $x_e \in \{0, 1, 2\}$ equal to the number of red copies. The actual colour degrees are $R(v) = \sum_{e \ni v} x_e$ and $B(v) = 2d_G(v) - R(v)$. A comparison is required in a colour precisely when that colour is present on the edge. Requiring both comparisons on every edge is a stronger sufficient condition that some, but not all, of our constructions satisfy.

The original conjecture and results for bipartite graphs appear in [1]; cacti are treated in [2]. The regular-graph result, a three-colour bound for subcubic graphs, and special subcubic families are established in [3]. In its arXiv version, Theorems 14–15 cover independent degree-three vertices and subdivisions of selected cubic edges into paths of length at least five. We give a full subcubic proof, retaining its complete finite interfaces. The later sections address different questions: unbounded-degree constructions, prescribed boundaries, target feasibility, and limitations of proposed proof methods.

The identifiers A1–A9, B1–B4 and C1–C5 refer to a prior inventory of research notes; they are preserved for traceability in Table 1. They are groups of related assertions, not a count of mutually independent or historically new theorems. We fold ordinary existence results for smaller subcubic families into A1 and retain stronger boundary statements separately. D1–D5 identify completed extensions documented after 17:00 China Standard Time on 27 October 2026, with their source snapshot fixed at 20:31:04 that evening. E1–E4 contain four completed notes from 28 October 2026, frozen at 10:25:33 China Standard Time. They add integer rather than merely modular separation, path/cycle completions, and parity-port certificates. Work completed after that cutoff, including the later component-dynamic-programming note, is outside this fixed snapshot. The original notes, snapshot hashes and a detailed coverage ledger accompany the manuscript.

Table 1: Coverage of the retained inventory.

Identifier	Assertion group	Location
A1	All subcubic graphs; safe interfaces and the two-mark kernel	Sections 2–9, Appendix A
A2	Three sparse direct-branch hypotheses	Section 12
A3	Degree-homogeneous, three-colourable direct cores	Section 13
A4–A5	Balanced half-edges and simultaneous preserved-colour extensions	Section 14
A6	Near-centre parity realisation and degree-gap separation	Section 15
A7–A8	Arbitrary clique root sums, block trees, and capacity extensions	Section 16
A9	Tree gluing of given coloured pieces	Section 17

Identifier	Assertion group	Location
B1	Trees, paths, decorated saturation, preserved subdivisions, offsets	Sections 10, 11, 10.1, 20.2
B2	Fixed targets, matching, factors, exchanges, defect bounds	Section 18
B3	Exact polynomial, free parameters, permanent and parity formula	Section 20
B4	Normal forms, peripheral elimination and minimal counterexamples	Section 19
C1	Cut, parity, modulo-three and modulo-two limitations	Sections 20.4, 25
C2	Nonincreasing single-edge and two-unit descent barriers	Section 26
C3	Target obstructions, nonconvexity and unbounded parity radius	Sections 27, 27.3
C4	Sharp fixed-interface path and extension thresholds	Sections 19.4, 16.1, 28
C5	Gluing, prescribed root sums and multiple marked ports	Sections 17.1, 28
D1	High minimum degree relative to chromatic number	Section 21
D2	Multiple CRT regions, low-side ports, matched tree attachments	Sections 21.2, 21.3
D3	Dense antitone targets: $8\delta \geq 7n + 6$	Section 22
D4	Complement lifts, nonregular families, structural restrictions	Section 24
D5	Dense forbidden targets and sequential region absorption	Section 23
E1	Signed gaps with two occurrences; paired partitions and fixed-mixed integer repairs	Section 29
E2	Degree-three/four path–cycle side against a covering matching	Section 30
E3	Protected parity joins and exact independent-side completions	Section 31
E4	Arbitrary-degree odd-component root ports and capacity-aware parity test	Section 32

The matching reductions, Tutte's factor criterion, jump-system exchange principles and polynomial interpolation are classical tools, not claims of invention. We give the required applications and elementary arguments so that the boundary between a feasible target and a legal colouring is explicit. The degree-constrained subgraph theorem used for D1 is quoted with its hypotheses and applied only to a simple host graph. The subcubic proof uses the degree-choosability theorem recorded in [4]. Other background dependencies are indicated where used.

This is a research preprint with extensive computer assistance. Its infinite-family assertions are justified by the displayed arguments and, where indicated, complete finite kernels plus lifting inductions. Random and bounded-order checks test implementations; they do not prove the conjecture. No independent expert review or proof-assistant verification is claimed, and no absolute historical-priority claim is made. In particular, none of the strategy obstructions below is a counterexample to the original conjecture.

2 The subcubic theorem

Theorem 2.1. *Let G be a connected simple graph with $G \ncong K_2$ and $\Delta(G) \le 3$. Then $\mathrm{lir}(^2G) \le 2$.*

The proof uses finite calculations for three auxiliary state systems. Their complete domains, recurrences and checking procedures are specified below and in Appendix A. The finite calculations are not enumerations of all graphs up to an order bound. Inductions on paths, trees and blocks extend the verified relations to graphs of arbitrary order. The only external colouring theorem used is the classical degree-list theorem, stated as Theorem A in [4].

3 Edge weights and path interfaces

Assign to every original edge e a weight $x_e \in \{0, 1, 2\}$, the number of its red copies. Define

$$R(v) = \sum_{e \ni v} x_e, \qquad B(v) = 2d_G(v) - R(v). \tag{1}$$

The required condition on an edge uv of weight x is

$$x > 0 \Longrightarrow R(u) \neq R(v), \qquad x < 2 \Longrightarrow B(u) \neq B(v). \tag{2}$$

When both endpoints have degree three, it is sufficient and necessary that their red degrees differ. The same observation applies to two degree-two vertices, with $B = 4 - R$ instead of $6 - R$.

Let $W = \{v : d_G(v) = 3\}$ and $H = G[W]$. Vertices of W are called *roots*; a connected component of H is called a *core*. Every edge outside H belongs to a maximal path with internal vertices of degree two. Such a path is an *ordinary connection* if its endpoints are distinct roots, a *return path* if both endpoints are the same root, and a *pendant path* if one endpoint is a root and the other is a leaf. An ordinary connection has length at least two, and a return path has length at least three. A pendant path can have any positive length. Distinct roots of an ordinary connection may belong to the same core. A length-one edge between roots is already an edge of H.

The weight of the first edge at a root is the corresponding *port*. A port is *pure* if its weight is 0 or 2, and *mixed* if its weight is 1. A root red degree is physically compatible with a port a when $a \le R \le a + 4$. At a return path its two port contributions must both be included in the same root degree.

Lemma 3.1 (Path interfaces)**.** *The following statements hold for physically compatible root degrees.*

1. *A two-edge ordinary connection with pure end ports a, b is admissible exactly when $R(u) \neq 2+b$ and $R(v) \neq 2 + a$.*

2. *A three-edge ordinary connection with arbitrary ports a, b is admissible exactly when*

$$a \neq b, \qquad \neg\big((a = 1 \text{ or } b = 1) \text{ and } R(u) = R(v) = 3\big). \tag{3}$$

3. *Opposite pure ports are admissible on every ordinary connection of length at least three. Any two pure ports are admissible at length at least four.*

4. *A return path accepts ports $(0, 2)$ at every legal length. If its root red degree is nonzero modulo three, each of $(0, 1)$ and $(1, 2)$ is also admissible.*

5. *A pendant path of length at least three accepts either pure root port. At length one, ports 0 and 2 exclude root red degrees 4 and 2, respectively; at length two they exclude 2 and 4, respectively.*

Proof. For a path with weights $x_1, \dots, x_\ell$, all comparisons between its internal degree-two vertices are equivalent to $x_i \neq x_{i+2}$. At a degree-three root of red degree r, its incident port a permits the next weight to lie in

$$P_r(a) = \{0,1,2\} \setminus (\{r-a : a > 0\} \cup \{r-a-2 : a < 2\}), \tag{4}$$

where values outside $\{0,1,2\}$ have no effect. Part 1 follows by direct substitution. For three edges, the middle comparison is $a \neq b$ and the remaining weight must lie in $P_{R(u)}(a) \cap P_{R(v)}(b)$. A pure port has at least two choices. A mixed port has at least two choices except at red degree 3, where its set is $\{1\}$. A pure-port set excludes 1 exactly at red degree 3. This proves part 2.

For longer paths, separate the odd and even edge positions. They form two three-colour paths under the relations $x_i \neq x_{i+2}$. At even length at least four, each position path has one fixed endpoint and an endpoint list of size at least two. At odd length at least five, one position path has two fixed endpoints at distance at least two and the other has two lists of size at least two. Both can always be coloured. Together with part 2 this proves part 3. If one port is mixed but its root red degree is not 3, its set in (4) again has at least two elements, so the same argument applies to longer paths.

For a return path set both endpoint red degrees equal to the actual root degree. Its nonzero residue excludes red degree 3; the preceding argument gives part 4. At a pure leaf port 0 or 2, the preceding weight has allowed set $\{0,1\}$ or $\{1,2\}$, respectively. The position-path argument proves part 5 for length at least four. At length three choose the leaf port opposite to the root port and intersect two sets of size at least two. The length-one and length-two exclusions follow directly from (2). □

We will use the safe set

$$\mathcal{R} = \{0,1,3,5,6\}. \tag{5}$$

In particular, pure ports at safe roots automatically satisfy either end of every two-edge connection. To attach a pendant path at a modular root of target residue ϕ, take port 2 exactly when its length is one and $\phi = 1$, or its length is two and $\phi = 2$; take port 0 otherwise. Both choices are allowed at any length when $\phi = 0$.

If W is empty, G is a single vertex, a path, or a cycle. A two-edge path takes weights $(0,0)$; longer paths take first weight 0 and last weight 2 and use the position-path argument with leaf endpoint conditions. For a cycle, give its edge positions a proper three-colouring in the auxiliary graph joining positions i and $i+2$ modulo its length. That auxiliary graph has maximum degree two. The resulting weights satisfy all comparisons. Also, for K_4, on edges $(01, 02, 03, 12, 13, 23)$ take weights $(1,2,0,0,0,2)$; the red degrees are $(3,1,4,2)$. We henceforth exclude these cases.

4 Three finite auxiliary lemmas

A core C has $3 - d_C(v)$ private ports at each vertex v. In an auxiliary port assignment its red degree is the sum of its core edge weights and its private port weights. Private ports here are free variables, not prescribed boundary weights.

Lemma 4.1 (Free safe ports)**.** *Suppose C is a tree, a cactus all of whose cycle blocks are odd, or a connected bipartite subcubic graph with at most two vertices of core degree three. There is an assignment for which all private ports are pure, every red degree belongs to $\mathcal{R}$, and adjacent core vertices have distinct red degrees.*

Lemma 4.2 (One mixed port)**.** *Let C be a bipartite subcubic cactus. Any specified private port can be set to 1 so that its root red degree is 1 or 5, all other private ports are pure, every root red degree is safe, and adjacent core vertices have distinct red degrees.*

We specify the finite calculations and their infinite extensions before stating the third lemma. A rooted state (x, r) records the incoming core-edge weight (or one actual free port at the final root) and the red degree of that vertex. Its domain is

$$\Omega = \{(x, r) : x \in \{0, 1, 2\},\ r \in \mathcal{R},\ x \le r \le x + 4\}, \qquad |\Omega| = 9. \tag{6}$$

The six rooted message sets are

$$\begin{aligned}
\mathcal{L} &= \{(0,0), (1,1), (1,3), (1,5), (2,6)\},\\
\mathcal{O} &= \{(0,1), (0,3), (1,1), (1,3), (1,5), (2,3), (2,5)\},\\
\mathcal{A} &= \Omega,\\
\mathcal{C} &= \{(0,0), (0,1), (1,3), (2,5), (2,6)\},\\
\mathcal{D} &= \{(0,0), (0,1), (0,3), (1,3), (2,3), (2,5), (2,6)\},\\
\mathcal{E} &= \{(0,0), (0,1), (1,1), (1,3), (1,5), (2,5), (2,6)\}.
\end{aligned}$$

For $c \in \{0, 1, 2\}$ child sets $X_1, \dots, X_c$, define

$$N(X_1, \dots, X_c) = \left\{ \left(x, x + \sum_{i=1}^{c} y_i + \sum_{j=1}^{2-c} p_j \right) : \begin{array}{l} (y_i, s_i) \in X_i,\ p_j \in \{0, 2\},\\ r = x + \sum_i y_i + \sum_j p_j \in \mathcal{R},\ r \ne s_i\ (1 \le i \le c) \end{array} \right\}. \tag{7}$$

Here $x \in \{0, 1, 2\}$. Every one of the $1 + 6 + 36 = 43$ ordered choices has one of the six message sets as its exact result.

For a non-anchor vertex of a cycle, its third incidence is a free pure port, denoted F, or one child message set X. Define a Boolean matrix M_X on Ω by

$$(M_X)_{(x,r),(y,s)} = 1 \quad \Longleftrightarrow \quad s \ne r \text{ and } \begin{cases} s - x - y \in \{0, 2\}, & X = F,\\ \exists (p, t) \in X :\ s = x + y + p,\ s \ne t, & X \ne F. \end{cases} \tag{8}$$

The rooted interface of a cycle product M consists of the states (z, r) for which there are $a, b \in \{0, 1, 2\}$ and $(b, s) \in \Omega$ with $r = z + a + b$, $s \ne r$, and $M_{(a,r),(b,s)} = 1$.

Proposition 4.3 (Finite closure identities)**.** *The node recurrence (7) is closed on the six message sets. Let $\mathcal{P}_k$ be the set of all products of k letter matrices from (8). Then*

$$\begin{aligned}
(|\mathcal{P}_2|, |\mathcal{P}_4|, |\mathcal{P}_6|) &= (14, 11, 2), & \mathcal{P}_8 &= \mathcal{P}_6,\\
(|\mathcal{P}_3|, |\mathcal{P}_5|, |\mathcal{P}_7|) &= (23, 6, 2), & \mathcal{P}_9 &= \mathcal{P}_7.
\end{aligned}$$

Their rooted interface types have the following multiplicities:

Parity of cycle length	*Number of letters*	*Rooted types*
odd	*2*	$\mathcal{C}$ *once,* $\mathcal{D}$ *thirteen times*
odd	*4*	$\mathcal{A}$ *ten times,* $\mathcal{D}$ *once*
odd	*6*	$\mathcal{A}$ *twice*
even	*3*	$\mathcal{A}$ *twenty-one times,* $\mathcal{O}$ *twice*
even	*5*	$\mathcal{A}$ *six times*
even	*7*	$\mathcal{A}$ *twice*

Moreover the node recurrence restricted to $\mathcal{L}, \mathcal{O}, \mathcal{A}$ stays within these three types.

Computer-assisted finite calculation. Evaluate (7) for the 43 choices. Construct the seven matrices by (8) and multiply them as Boolean relations. The files `safe_cactus_certificate.py` and `safe_all_cactus_certificate.py` independently recompute the complete layers, compare sets rather than just their cardinalities, and compute every rooted interface. They use Python integer and set operations only. The complete layer sets are supplied in `safe_cactus_interfaces.json` and `safe_all_cactus_interfaces.json`. Appendix A gives the common execution command. □

Proof of the tree and cactus parts of Lemmas 4.1 and 4.2. For a tree, root at a vertex with an available private port, use that port as the incoming incidence, and apply (7) upward. Only $\mathcal{L}, \mathcal{O}, \mathcal{A}$ occur. All accept incoming weight 0, which gives the free-port assertion, and incoming weight 1 with red degree 1 or 5, which gives the marked assertion.

For a cactus of maximum degree three, different cycle blocks cannot share a vertex. Contract each cycle block to a unit and induct on the resulting tree of units. A non-anchor cycle vertex has precisely the third-incidence choices in (8); two children there would violate the degree bound. A cycle of length ℓ uses $\ell - 1$ letters. The identities in Proposition 4.3 imply, by induction under multiplication by any two letters, that every longer product of the same parity has one of the listed rooted types. Thus the argument covers arbitrary cycle lengths and numbers of blocks.

Every one of the six types accepts incoming weight 0. This proves the odd-cactus assertion. In a bipartite cactus every cycle is even, and the induction stays in $\mathcal{L}, \mathcal{O}, \mathcal{A}$. Root at the specified private port and choose $(1, 1)$ or $(1, 5)$ to prove Lemma 4.2. The incoming port is an actual private incidence, not an extra edge added to the original degree. □

For core chains with one pure private port at each internal vertex, write

$$(x, r)\ T\ (y, s) \quad \Longleftrightarrow \quad r \neq s, \quad s = x + y + p \in \mathcal{R} \text{ for some } p \in \{0, 2\}. \tag{9}$$

Both states lie in Ω. The supplied certificate gives a six-step walk for every ordered pair of states, so

$$T^6 = \Omega \times \Omega. \tag{10}$$

Every state has an incoming neighbour, and therefore $T^k = \Omega \times \Omega$ for every $k \geq 6$. Here internal chain vertices have original degree three, unlike the degree-two internal vertices in Section 3.

Proof of the remaining part of Lemma 4.1. If there are no core-degree-three vertices, the core is a single vertex, a path, or an even cycle, already covered above. Otherwise compress its core-degree-two chains. If n_1, n_3 count core degrees one and three, and $\beta = |E(C)| - |V(C)| + 1$, then

$$n_1 = n_3 + 2 - 2\beta. \tag{11}$$

With one degree-three vertex the only shapes are a three-arm tree and an even cycle with an arm. With two, the only shapes are a double-star tree, an even cycle attached to a fork with two arms, an even cycle with an arm at each of its two forks, a theta, or two even cycles joined by a chain. These follow by counting loops and paths in the compressed shape, whose two forks have degree three.

There are 163 complete finite shape and length representatives, with their domain specified in Table 3. Every representative has a witness assigning odd safe red degree to each fork, consistent with its three incident weights. The checker `sparse_bip_core_certificate.py` verifies the complete domain, all fork sums and all chain relations, as well as (10).

Keep a length $n \le 6$, replace a longer odd length by 7, and a longer even length by 8. Any chain of length at least seven has at least six internal transitions, so (10) lifts its witness with unchanged end weights and fork degrees. Parities and simplicity restrictions are preserved. An arm can have arbitrary positive length: an odd-red fork can be completed immediately at a leaf by two pure ports; from every state one step reaches a state that admits such a completion, and every state has an outgoing step. These finite assertions are also checked. Thus all arm lengths and all chain lengths are covered. □

Definition 4.4 (Four-port tree)**.** A four-port tree is a subcubic tree core with two distinguished vertices a, b at odd distance. Neither has a short port. On each side of the a–b path there are exactly two positions, each either a direct long private port at the distinguished vertex or an arm ending at a leaf with one long and one short port. Every other internal vertex of such an arm or of the a–b path has one short private port. Thus there are precisely four long ports. The distinguished vertices have no other incidences.

Figure 1 shows one possible topology. A position at a distinguished vertex may instead be a direct long port.

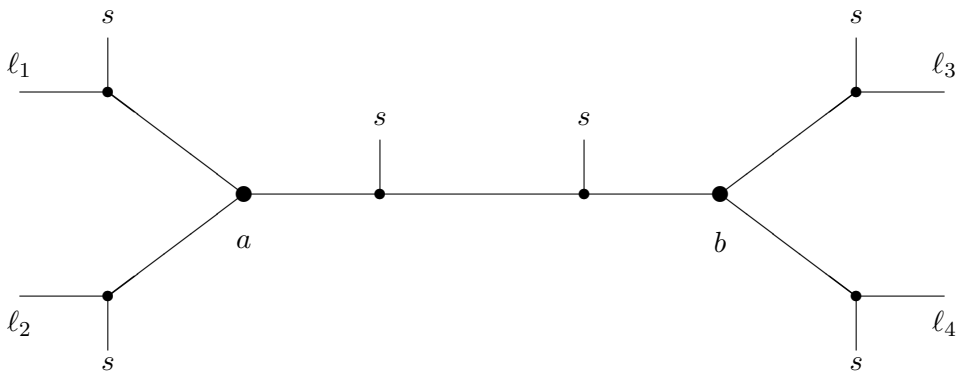


Figure 1: A four-port tree with two distinguished vertices a, b, four long ports ℓ_i, and short ports labelled s. The middle path is odd; the arms and middle path may be longer than shown, with one short port at every added internal core vertex. All vertices drawn have original degree three. Only a, b have no short incidence and may use red degrees 2 or 4.

Lemma 4.5 (Two mixed ports)**.** *In any four-port tree, any two distinct long ports can be set to 1 so that their roots have red degree different from 3. All other private ports can be pure, adjacent core vertices can have distinct red degrees, and every vertex carrying a short port can have safe red degree. The two distinguished vertices may have red degree 2 or 4.*

Proof. For an arm, its incoming message at the leaf is $\mathcal{L}$ if the long port is pure, and $Q = \{(0,1),(2,5)\}$ if the long port is marked. At an internal vertex the message map is

$$U(X) = \{(x, x+y+p) : (y,s) \in X,\ p \in \{0,2\},\ x+y+p \in \mathcal{R},\ x+y+p \neq s\}. \tag{12}$$

Exact iteration gives cardinalities $5, 7, 9, 9$ from $\mathcal{L}$, and $2, 3, 5, 7, 7, 9, 9$ from Q. In each case the last two sets are equal to Ω, not merely equinumerous. Thus an unmarked position has four types, direct or arm lengths $1, 2, \ge 3$; a marked position has seven types, direct or arm lengths $1, 2, 3, 4, 5, \ge 6$.

Up to exchanging the two ends, the mark distribution is $(2,0)$ or $(1,1)$. The first has $\binom{8}{2}\binom{5}{2} = 280$ choices of unordered pairs of position types; the second has $(7 \cdot 4)^2 = 784$ choices. Two direct

marked ports at the same distinguished vertex are included. The connecting path is odd, and its representatives have lengths $1, 3, 5, 7$. Hence the complete domain contains $(280 + 784) \cdot 4 = 4256$ cases.

For each case, `weak_two_marked_interfaces.json` gives the end states, the two position choices at each end, and compatibility with the connecting path. The checker recomputes the domain, arm sets, actual sums and connecting relations; every case passes. A second verifier recomputes existence from these definitions without using the stored witnesses. Details appear in Appendix A.

Arm fixed points prove lifting at every larger arm length. For the middle path, (10) proves lifting at every length at least seven if the two end degrees are safe. If an end red degree is 2 or 4, it differs from every internal safe red degree; a word valid with the same end-edge weight and virtual end red degree 3 is still valid with the actual degree. The virtual state is physical for every possible end weight. Thus larger odd lengths preserve both actual end degrees and end weights. These observations prove the lemma for arbitrary four-port trees, rather than only for the finite representatives. □

5 Classifying cores and counting long ports

Call an ordinary connection *short* if its length is two, and *long* if its length is at least three. For a bipartite core C with parts A, B, count the following objects on each part:

1. every pendant port rooted in that part;
2. every endpoint in that part of a short connection whose two roots both belong to C;
3. every vertex in that part incident with neither a pendant path nor any short connection, including short connections to other cores.

Let the resulting counts be k_A, k_B. In item 3 a core-degree-three vertex is always counted, even though it has no private port.

Assign cores to classes S and M by the following fixed rules, before choosing any weights. Put in M every core with a return path, every nonbipartite core incident with an ordinary three-edge connection, and every bipartite core incident with an ordinary three-edge connection and satisfying $\max(k_A, k_B) \geq 2$. Among the remaining cores, put in S those covered by Lemma 4.1, and put the rest in M.

The letters specify how a core will be handled, not that all degrees in its final colouring must have a common residue. In particular, Lemma 4.5 may give red degree 2 or 4 at vertices of an S core that carry no short port.

Lemma 5.1. *An* S *core incident with an ordinary three-edge connection is bipartite and has at most four long ports. If it has four ports that belong to three-edge connections between* S *cores, it is a four-port tree as in Definition 4.4.*

Proof. Nonbipartite cores incident with such a connection have been assigned to M. For a remaining bipartite core, $k_A, k_B \leq 1$. Let p be the number of pendant ports, s the number of short connections internal to C, and t the number of vertices in item 3. Then

$$t + p + 2s = k_A + k_B \leq 2. \tag{13}$$

There is no return path. A core leaf has at most one long port, plus one more if it is counted in t. A core-degree-two vertex with a long port is counted in t. A core-degree-three vertex is counted in

t but has no long port. Therefore, writing L for the total number of long ports, (11) gives, for a nontrivial core,

$$L \le n_1 + t - n_3 = 2 - 2\beta + t \le 4 - 2\beta - p - 2s \le 4. \tag{14}$$

A one-vertex core has only three ports.

If four of these ports belong to the stated three-edge connections, all inequalities are equalities. Thus $\beta = p = s = 0$, $t = 2$, and the two vertices counted in t lie in opposite parts. All core-degree-three vertices are among them. Their connecting path has odd length. Each other internal core vertex has its one short port; each other core leaf has one long and one short port. Removing the middle path leaves exactly two positions at each end, either direct ports or unbranched arms. There are no other long ports, pendant paths or internal short connections. This is precisely Definition 4.4. □

Corollary 5.2. *Every* S *core incident with an ordinary three-edge connection is a bipartite cactus.*

Proof. Its L is positive, so (14) gives $\beta \le 1$. A connected graph with at most one cycle is a cactus. □

6 Making all safe three-edge connections compatible

Form an indexed multigraph F whose vertices are the S cores. Every ordinary three-edge connection with both ends in S gives an edge of F. Roots in the same core give a loop, and different connections may give parallel edges. Loops contribute two to degree. By Lemma 5.1, $\Delta(F) \le 4$.

Lemma 6.1. *There is a spanning forest T of F and an orientation of $D = E(F) \setminus E(T)$ with outdegree at most two at each vertex. Outdegree two is possible only at a vertex of F of degree four.*

Proof. Take any spanning forest. Orient D so that its outdegree at each vertex lies between the floor and ceiling of half its degree. Such an orientation is obtained by joining each odd-degree vertex to a temporary vertex, orienting Euler tours in the resulting even-degree multigraph, and deleting the temporary edges. A loop has one tail.

In a component of F with at least two vertices, every vertex has an incident forest edge. Thus its D-degree is at most three, and at most two if its F-degree is at most three. In a singleton component all edges are loops, and at most two loops are possible. These facts prove both assertions. □

For every S core of outdegree zero use Lemma 4.1. For outdegree one, mark its outgoing half-port by 1 and use Corollary 5.2 and Lemma 4.2. For outdegree two, Lemma 5.1 gives a four-port tree, to which Lemma 4.5 applies. Only long ports are marked. Every short-incident root remains safe.

Each D edge now has exactly one mixed and one pure port, with red degree different from 3 at its mixed end. It is compatible by Lemma 3.1. A loop in F uses two distinct actual roots of the same core; it is not an original return path.

Every T edge still has pure ports a, b. Choose a colour-complement bit η_C for each core. Complementation changes every core weight and private port from x to $2 - x$ and changes each root red degree from r to $6 - r$. To make the two ports of a T edge opposite impose

$$\eta_C \oplus \eta_{C'} = 1 \oplus (a/2) \oplus (b/2). \tag{15}$$

These equations are simultaneously solvable on a forest. Complementation preserves safety at short ports, unequal red degrees on core edges, and the condition that a marked root is not of red degree 3. Thus all D connections remain compatible.

All ordinary three-edge connections between S cores are now compatible. At an S–M three-edge connection the S port is pure, since marking was confined to D. Set the M port opposite to it. At a three-edge connection between two M roots use ports 0 and 2. At longer ordinary connections any M port can be set to 0. These choices put no restriction on the eventual physical red degrees of M roots.

7 Targets and realizability modulo three

All arithmetic on target values in this section is in $\mathbb{F}_3$.

Lemma 7.1 (Realizing sums)**.** *Let C be a connected simple graph and $q : V(C) \to \mathbb{F}_3$. There are weights in $\{0, 1, 2\}$ whose incident sums equal q modulo three if and only if, in the bipartite case with parts A, B,*

$$\sum_A q - \sum_B q = 0. \tag{16}$$

There is no constraint if C is nonbipartite.

Proof. In a bipartite graph, each edge contributes once to each part, proving necessity. Fix a spanning tree. Setting non-tree weights to zero, remove leaves successively and set their parent edge to the remaining target. The final equation is exactly (16). For a nonbipartite graph, sign vertices alternately along the tree. Some non-tree edge has equal signs at its ends. Its coefficient in the signed defect is 2 or -2, invertible in $\mathbb{F}_3$. Choose its weight to cancel the defect and then eliminate along the tree. Choose representatives 0,1,2 for the resulting field elements. □

For an M core, let $c(v)$ be the sum of its private ports and seek a proper target colouring $\phi(v) \in \mathbb{F}_3$ of its core edges, then realize the core sums $q(v) = \phi(v) - c(v)$. The final actual degree satisfies $R(v) \equiv \phi(v)$, so adjacent core degrees differ.

If a short connection joins v to an S root with port a, delete $(2 + a) \bmod 3$ from the list at v. Denote the resulting list by $L(v)$. Since a is pure, only colours 1 or 2 are deleted; hence

$$0 \in L(v), \qquad |L(v)| \geq d_C(v). \tag{17}$$

Indeed at most $3 - d_C(v)$ private ports can impose deletions. A return path or a three-edge port at a vertex leaves a strict surplus $|L(v)| > d_C(v)$ there.

We use the following established theorem [4, Theorem A]: a connected graph that is not a Gallai tree is colourable from any lists of sizes at least the vertex degrees. A Gallai tree is a connected graph whose blocks are complete graphs or odd cycles. Also, a single strict-surplus vertex suffices in any connected graph: root a spanning tree there and colour in reverse tree order.

7.1 Nonbipartite cores

If a nonbipartite M core has a return path or an ordinary three-edge connection, it has a strict-surplus vertex. Otherwise, if it were a Gallai tree, its blocks would be bridges, odd cycles, or K_4. A K_4 uses all degree-three incidences, forcing $G = K_4$, already treated. Without it the core is an odd-cycle cactus and would have been assigned to S. Thus the remaining core is not a Gallai tree. In all cases a proper list target exists, and Lemma 7.1 realizes it for any subsequent private offsets. No bipartite defect needs to be balanced.

7.2 Bipartite cores with at least two safe short exits

Let f count short connections from C to S cores, with multiplicity. If $f \geq 2$, choose a proper target from the lists. This is justified as follows. If C contains a cycle, its bipartite structure makes it a non-Gallai graph. If it is a tree, its assignment to M implies the presence of a return path or a three-edge connection, giving a strict-surplus vertex.

Reserve this core's own port on each of two safe short exits. Each can be set independently to 0 or 2: the opposite root is safe and therefore accepts either value. Each changes the defect in (16) by a nonzero element of $\mathbb{F}_3$, without changing any target. Two such binary changes cover all residues and will balance the defect after other offsets have been fixed.

7.3 Bipartite cores with at most one safe short exit

Suppose $f \leq 1$. If C has a return path, choose its root in part B, set $\phi = 0$ on A and $\phi = t$ on B, and choose $t \in \{1, 2\}$ to respect the possible single list restriction. The return path accepts all three port pairs $(0, 1), (0, 2), (1, 2)$, because its root target is nonzero. Their total contributions $1, 2, 3$ cover the three defects.

Now suppose there is no return path. If C was assigned to M by the k-count rule, some side has at least two objects counted in k. Any other bipartite M core not already covered has at least three core-degree-three vertices; otherwise Lemma 4.1 would have placed it in S. One side therefore has two such vertices, each counted by item 3. Again one side has at least two counted objects.

Take this side as A, put $\phi = 0$ on A, and put $\phi = t \in \{1, 2\}$ on B. Since all lists contain 0 and there is at most one deletion in total, a choice of t respects all lists. Two objects on A will provide independent defect adjustments.

8 Balancing all cores without disturbing short connections

First fix the target at every root incident with a short connection. For a short connection between two M roots, give either endpoint the port

$$p = \begin{cases} 2, & \text{the other root has target 2,} \\ 0, & \text{otherwise.} \end{cases} \tag{18}$$

Then the other root target differs from $(2 + p) \bmod 3$, as required by Lemma 3.1. Give each M pendant its port from Section 3; give each return path the default pair $(0, 2)$ until a selected return adjustment is made.

Consider the remaining case from Section 7: a bipartite core with targets 0 on A, t on B and two counted objects on A. Define

$$\delta_C = \sum_{v \in A} (\phi(v) - c(v)) - \sum_{v \in B} (\phi(v) - c(v)). \tag{19}$$

Its counted objects have the following interpretations.

1. A pendant port at a zero-target root may take either pure value, at every length, by Lemma 3.1.

2. For an internal short connection whose endpoint $u \in A$ has target 0, the *other endpoint's own port* is free to take 0 or 2. The actual red degree at u is 0 modulo three, hence avoids both 2 and 4. The comparison at the other endpoint is protected by the already chosen port at u in (18).

3. A zero-target root with no pendant and no short incidence may change its target from 0 to $3 - t$. This remains different from all neighbouring core targets t. No short port or pendant choice depends on this target.

Each object changes (19) by a nonzero element of $\mathbb{F}_3$. The variable freed by item 2 is at the opposite endpoint, so its sign in the defect can be either sign; nonzeroness is all that is needed. Different short endpoints free different actual ports. If both ends are on A, both actual ports are independently free. An item 3 vertex has no short or pendant incidence, so it cannot overlap variables of the first two kinds. Thus two counted objects really give independent binary choices.

For any nonzero $\alpha, \beta \in \mathbb{F}_3$,

$$\{\alpha\epsilon + \beta\epsilon' : \epsilon, \epsilon' \in \{0, 1\}\} = \mathbb{F}_3. \tag{20}$$

Consequently the two chosen objects can cancel any initial defect. The two-safe-exit case uses the same calculation. The return case uses its three total contributions instead.

All cores can be balanced separately. Changing a target in item 3 never changes a short endpoint or pendant target. Changing a port uses only a pendant, a short connection internal to the same core, or this core's end of a connection to an S root. It never changes another M core's offsets. On a safe short exit the opposite root accepts either pure port, and its own degree does not change. Long path interfaces accept arbitrary actual root degrees. There is therefore no remaining coupled system between different modular cores.

Apply Lemma 7.1 to each core with its balanced targets and final offsets. All resulting core weights belong to $\{0, 1, 2\}$. The actual root degrees are the sums of these incident core weights and the fixed ports, so they are physical and satisfy $R(v) \equiv \phi(v)$. Every direct core edge has distinct red degrees at its ends and hence distinct blue degrees as well.

9 Completion of the proof

Proof of Theorem 2.1. The cases with no degree-three root and the case K_4 were treated in Section 3. Otherwise decompose the graph into cores and paths and classify its cores as in Section 5. Sections 6–8 assign all core weights and all root ports, with unequal degrees along every direct core edge. It remains to fill the path interiors.

For a short connection between two S roots, both roots carrying short ports are safe. Any exceptional red degree 2 or 4 allowed by Lemma 4.5 occurs only at a root without short ports. For an S–M short connection, the safe root's comparison is automatic and the modular root's comparison follows from its list. For an M–M short connection, (18) and the frozen short targets ensure both comparisons; Section 8 preserves them.

Every ordinary three-edge connection either has opposite pure ports or is a D edge with one mixed port whose root degree is not 3. Every longer ordinary connection has pure ports. Return and pendant paths have their admissible pairs or ports. Lemma 3.1 therefore fills every path.

The interiors are edge-disjoint, and their first and last weights were already counted in the root degrees. Filling them does not alter a root degree. This verifies core edges, internal path edges, root-end edges and leaf-end edges, exhausting $E(G)$. Replacing a weight x by x red and $2 - x$ blue copies yields (2) on every edge of the doubled graph. □

The argument uses the common identity $B = 6 - R$ at the roots, their three-incidence budget and the resulting bound $\Delta(F) \leq 4$. No theorem for graphs of larger maximum degree is asserted here.

10 Interfaces with arbitrary endpoint degrees

Here a *root* means a vertex of original degree at least three, without an upper bound. We use the same decomposition into direct root edges, ordinary paths, return paths and pendant paths as before. A non-direct ordinary path has two different actual end edges; their weights may be chosen independently. A direct edge has one shared weight. This distinction is essential throughout.

For a path $x_1 = a, \dots, x_\ell = b$ with internal vertices of degree two, prescribed final endpoint counts (r, s) give the allowed next weights

$$P_{r,s}(a) = \{0,1,2\} \setminus \big(\{r-a : a > 0\} \cup \{4-a-s : a < 2\}\big). \tag{21}$$

Values outside $\{0,1,2\}$ have no effect. Every internal comparison is equivalent to $x_i \neq x_{i+2}$. Thus the odd and even positions form independent proper three-colour paths with endpoint constraints.

Lemma 10.1. *If the two allowed sets in (21) each have at least two elements, any end weights extend at length at least four; at length three they extend if and only if $a \neq b$. Opposite pure end weights extend at every length at least three for all physical endpoint counts. If all four endpoint colour degrees are at least four, distinct end weights extend already at length two.*

Proof. At length three choose the middle value in the intersection of two two-element subsets of a three-element set. The other condition is $a \neq b$. At length four choose $x_2 \in P_L \setminus \{b\}$ and $x_3 \in P_R \setminus \{a\}$. At length five choose $x_3 \notin \{a, b\}$ and different $x_2 \in P_L, x_4 \in P_R$. For greater lengths each position path has at least two edges between its endpoint constraints. A three-colour path of at least two edges connects arbitrary prescribed colours, by induction starting with a middle colour different from both ends. For a pure end weight, at most one value is forbidden, proving the opposite-pure assertion. If $r, s \geq 4$, a mixed port forbids no value, and a pure port can only forbid its own value, when the relevant colour degree is four. At length two distinct ports therefore satisfy both endpoint checks. □

We will also use the following small extension of the first part: at length at least three, different end weights extend if one allowed set is $\{1\}$, the other contains 1 and has at least two elements, and the opposite fixed end weight is pure. At lengths three and four this follows from the same explicit choices; at length five the two allowed even-position values can be made different; greater lengths follow from the position-path induction.

10.1 Exact preservation under subdivision

Theorem 10.2. *Let a legal colouring of ${}^2Q$ be given. Replace a set of original edges $e = uv$ by internally disjoint paths of lengths $L_e \geq 4$. Set*

$$\epsilon_e(v) = \mathbb{1}_{\{x_e=1,\ R_Q(v)=B_Q(v)\in\{1,3\}\}}.$$

If $L_e \geq 4 + \epsilon_e(u) + \epsilon_e(v)$ for every replaced edge, the colouring extends preserving every old vertex colour degree and every unreplaced edge weight. At lengths four and five this condition is also necessary when both new end weights must equal x_e.

Proof. Keep each end weight equal to $a = x_e$. Pure ports have at least two allowed next values. For $a = 1$, the forbidden next values are $R - 1$ and $3 - B$. If both belong to $\{0,1,2\}$, then $R, B \in [1,3]$. Since $R + B$ is even, the allowed set has size one exactly when $(R, B) = (1,1)$ or $(3,3)$, and that set is $\{1\}$. At length four each next value must differ from the opposite fixed end value a; one singleton is therefore fatal. At length five the two next values must differ from one another, so precisely

two singletons are fatal. At length at least six each position path can connect arbitrary nonempty boundary sets. These conditions give the stated threshold. All old incident contributions remain identical, so all unreplaced comparisons persist simultaneously. □

For cubic initial data there are 60 ordered legal end states: for each a, each endpoint red degree lies in $\{a, \dots, a+4\}$ and the two degrees differ. All 40 pure-port states extend at length four. Of the 20 mixed-port states, exactly eight fail: one end red degree is three and the other is one of $1, 2, 4, 5$, in either order. These are fixed-data obstructions, not counterexamples to Theorem 2.1.

In particular selected edges whose endpoints all have degree at least four can always be replaced by paths of length at least four while preserving a given colouring. Known regular-graph colourability [3] makes this unconditional for every connected regular mother graph of degree at least four. Known bipartite colourability [1] gives the same conclusion for a bipartite mother graph when every selected endpoint has degree at least four. The initial colouring is supplied by those theorems; it is not assumed for a general unknown mother graph.

If a vertex-minimal counterexample to the general conjecture exists, it has no ordinary degree-two thread of length at least five between distinct nonadjacent roots; if both roots have degree at least four, length at least four is excluded. Replace that thread by a single edge. The smaller graph is simple and is not K_2. Its legal colouring exists by minimality. On a mixed new edge at most one endpoint can have $(R, B) = (3, 3)$, since two such endpoints would already violate the edge. Theorem 10.2 restores the thread, a contradiction. The nonadjacency assumption cannot be dropped in this argument.

11 One-bridge trees and decorated paths

For a rooted tree (T, r) and an external bridge of weight c, let $P_c(T, r)$ be the achievable *internal* root red sums when all internal edges are legal after the root receives external contribution $(c, 2-c)$. The external bridge itself is not yet checked.

Lemma 11.1. *For a one-vertex tree all three sets are $\{0\}$. For a two-vertex tree, $P_0 = \{0\}, P_1 = \{0, 1, 2\}, P_2 = \{2\}$. Every tree of order at least two has $|P_1| \geq 3$; every tree of order at least three has $|P_0|, |P_2| \geq 2$.*

Proof. Use simultaneous induction on tree order. Delete the root, leaving k rooted subtrees with root internal degrees d_i. If the root edge to subtree i has weight a_i and its internal root sum is p, the only new restrictions, for final parent counts A, B, are

$$a_i > 0 \Rightarrow p + a_i \neq A, \qquad a_i < 2 \Rightarrow 2d_i - p + 2 - a_i \neq B.$$

Thus a mixed edge forbids at most two child candidates, a pure edge at most one. Colour complementation gives $P_{2-c} = \{2k - p : p \in P_c\}$.

For $c = 0$, put all $a_i = 1$. The internal sum k is feasible: larger children have three mixed candidates, and singleton children have counts $(1, 1)$, different from parent counts $(k, k+2)$ unless $k = 1$, which would be the two-vertex base case. Change one a_i to 2 to seek sum $k+1$ and parent counts $(k+1, k+1)$. That child must avoid $p = k-1$ in P_2. A child of order at least three has two candidates; a singleton fails only at the excluded $k = 1$; a two-vertex child fails only at $k = 3$. The only remaining configuration is three two-vertex children. Take root weights $(0, 1, 1)$ and all child internal weights zero instead. The parent is $(2, 6)$; the pure-blue child root has blue degree four and the mixed child roots have counts $(1, 3)$. This gives a second sum distinct from $k = 3$. Complementation proves the P_2 bound.

For $c = 1$, all $a_i = 1$ yield sum k and counts $(k+1, k+1)$. Changing one a_i to zero seeks sum $k-1$ and counts $(k, k+2)$; the child must avoid $p = 2d_i - k$ in P_0. The only exception not handled by the induction is two two-vertex children. In that case set both root weights and both child internal weights to zero, giving internal sum zero and parent counts $(1, 5)$, with child blue degrees four. Consequently a feasible sum $a \neq k$ always exists. Its colour complement $2k - a$, together with k, gives three distinct sums. □

For an outside vertex of final counts (A, B) the bridge imposes at most one or two further exclusions. Lemma 11.1 gives the complete compatibility table

Tree order	$c = 0$	$c = 1$	$c = 2$
1	$B \neq 2$	$A, B \neq 1$	$A \neq 2$
2	$B \neq 4$	always	$A \neq 4$
at least 3	always	always	always

Here "always" assumes physical counts containing the bridge copies. This is an interface statement; adding the bridge changes outside degrees and may invalidate other outside edges.

11.1 Saturation of decorated paths

Theorem 11.2. *Let every internal vertex of a path have at most one bridge to a disjoint rooted tree. A path of at least six edges accepts arbitrary physical final endpoint counts and arbitrary prescribed first and last edge weights, while colouring all attached trees. The endpoints may be the same vertex if both incident contributions are counted.*

Proof. Classify an internal vertex as A (bare), L (a singleton tree), S (a two-vertex tree), or F (a larger tree). With incoming and outgoing weights a, p, bridge weight c, and degree $D = 2$ or 3, its red sum is $r = a + p + c$. For A, put $c = 0$. The compatibility table above says: for L, $a + p = 0$ forces $c = 0$, $a + p = 4$ forces $c = 2$, and otherwise any c works; for S, $a + p = 2$ forces $c = 1$ and otherwise any c works; for F every c works.

Record a state (p, D, r). There are 24 possible states,

$$\mathcal{U} = \{(p, 2, r) : p \leq r \leq p + 2\} \cup \{(p, 3, r) : p \leq r \leq p + 4\}, \quad p \in \{0, 1, 2\}.$$

An old state with outgoing weight a can precede a new state (p, D', r') exactly when the indicated bridge is allowed and $a > 0 \Rightarrow r \neq r'$, $a < 2 \Rightarrow 2D - r \neq 2D' - r'$. Let T_X be the resulting monotone map on state subsets. Complete integer tables are given in Appendix B.1, including the four lower-bound families F_1, F_2, F_3, F_4 of sizes $30, 19, 7, 3$.

The initial endpoint has just 93 comparison types: each colour count is $0, \ldots, 6$ or greater than six, with physical and parity restrictions. For each type and first letter the reachable set contains an element of F_1. For every $M \in F_i$ and letter X, $T_X(M)$ contains an element of F_{i+1}, for $i = 1, 2, 3$. For each of the three elements of F_4, applying T_X gives the entire locally feasible state set Ω_X. These are complete finite identities, recomputed without the constructor by `decorated_path_certificate_check.py`.

It follows by monotonicity and induction that after five internal vertices the reachable set is Ω_X, and remains saturated after every further vertex. For a prescribed final weight, Ω_A has three possible red sums, Ω_L, Ω_F have five and Ω_S has at least four. The right boundary forbids at most two, so it can always be met. Restore the attached trees using the bridge lemma. The clipping of large endpoint counts is exact, since internal colour degrees never exceed six. This proves arbitrary length and arbitrary tree shapes from the finite identities. □

12 Sparse direct branch cores

Theorem 12.1 (A2)**.** *Let $W = \{v : d_G(v) \geq 3\}$ for a connected simple graph $G \neq K_2$. Each of the following hypotheses implies $\operatorname{lir}(^2G) \leq 2$:*

1. *$G[W]$ is independent;*
2. *there is no degree-three vertex and $\Delta(G[W]) \leq 1$;*
3. *$\Delta(G[W]) \leq 1$ and every non-direct ordinary branch path has length at least four.*

Return paths of length at least three and pendant paths of arbitrary positive length are allowed in all three statements.

12.1 Independent branch vertices

Proof of the first statement. For a root let l, h, p count return paths, ordinary incidences and pendant paths, respectively; its original degree is $D = 2l + h + p$. On W form a simple graph joining distinct roots connected by an ordinary path of length at least three. Choose a maximal independent set A, then a maximal independent set B in the remaining graph, and put $C = W \setminus (A \cup B)$. Every C root has a long neighbour in both A and B; every isolated root belongs to A.

Give every return path ports $(0, 2)$. Give pendant root ports 0, 2, 1 on A, B, C, respectively. On ordinary paths assign port pairs $AA : (0, 0)$, $BB : (2, 2)$, $AB : (0, 2)$, $AC : (0, 1)$, $BC : (2, 1)$. An AA or BB path necessarily has length two. Balance-orient the multigraph of CC paths and give each tail port 0 and head port 2. For a length-two AC path whose C root has degree three, change its C port to 2; for such a BC path change it to zero.

At a C root of degree at least four the only imbalance between its red sum and D is at most one from the balanced CC orientation. At degree three the two long A, B incidences leave at most one other incidence, so either one CC imbalance or one short-path correction occurs, again of magnitude at most one. If the red sum is $D - 1$, change its long A port from 1 to 2. If it is $D + 1$, change its long B port from 1 to zero. These independent end ports affect no other root. Now every C root has $(R, B) = (D, D)$.

At an A root the counts are $(2l, 2(D - l))$, and at a B root they are reversed. The larger count is an even integer at least four. Equality to four at a root with an external path occurs only when $D = 3, l = 1, h + p = 1$. Such a root is exceptional only in the following two cases. If its unique external path has length two and joins another A root, change its own port from 0 to 2; if both ends are exceptional change just one. Its counts become $(4, 2)$ and the midpoint has $(2, 2)$, legal on the red and blue sides. If its unique external path is a two-edge pendant, change that port to 1 and use the word $(1, 1)$; root, middle and leaf counts are $(3, 3), (2, 2), (1, 1)$. These exceptional roots have no long ordinary incidence, so none of the previous adjustments is disturbed.

The remaining two-edge ordinary checks are listed below. The displayed root counts refer to the colour present on the corresponding port.

Type	Ports	Middle (R, B)	Root comparisons
AA	$0, 0$	$0, 4$	both blue at least 6
BB	$2, 2$	$4, 0$	both red at least 6
AB	$0, 2$	$2, 2$	relevant counts at least 4
AC, $D_C = 3$	$0, 2$	$2, 2$	blue at least 4; red 3
AC, $D_C \geq 4$	$0, 1$	$1, 3$	blue at least 4; both at least 4
BC	complement	complement	same inequalities
CC	$0, 2$ or $2, 0$	$2, 2$	both root counts at least 3

For long ordinary paths the end ports differ. An A or B allowed set has at least two elements and contains 1. At a C root the only singleton occurs at $D = 3$ with port 1, when it is $\{1\}$; its other end is an A or B pure port. The general path lemma and its singleton extension therefore apply. Every return path has opposite pure ports. Pendant pure ports use the constant word at lengths one or two, with the unique bad two-edge case already repaired; at longer lengths use the opposite pure leaf port. A C pendant uses (1) or $(1, 1)$ at lengths one or two; otherwise give its leaf port zero and apply the singleton extension. All root counts were fixed first. Consequently the independent path fillings are simultaneously legal. □

12.2 A direct matching with no degree-three roots

Proof of the second statement. Compress all paths, using a private placeholder leaf for each pendant path only for the purpose of orientation. A balanced orientation of this multigraph gives outdegrees $o(v) \in \{\lfloor D/2 \rfloor, \lceil D/2 \rceil\}$. Give non-direct tail ports 1 and head ports 0; give a pendant root port 1 or 0 according to its direction; give each direct edge the shared weight 1. If $i_H(v)$ indicates an incoming direct edge, the initial counts are

$$R_0(v) = o(v) + i_H(v) \geq 2, \qquad B_0(v) = 2D - R_0(v) \geq 5.$$

Repair each direct matching edge independently. If only its red degrees coincide, change its weight to zero. The head initially had red degree at least three, so both new red degrees remain at least two. If only its blue degrees coincide, change its weight to 2. The tail initially had blue degree at least six, so both new blue degrees remain at least five. If neither coincides, retain weight 1.

If both coincide, orient the edge $u \to v$. Equal original degrees and $o(u) = o(v) + 1$ force $D = 2k + 1$, $k \geq 2$, with common initial counts $(k + 1, 3k + 1)$. There are k other outgoing incidences at u. If one is an ordinary or return path, change its tail port 1 to 2, giving u counts $(k + 2, 3k)$. Otherwise choose an outgoing pendant port and change 1 to zero, giving $(k, 3k + 2)$. The mixed direct edge is now legal. Only tail ports were modified; head ports stay zero, so even repairs selecting the same connection cannot interfere. Different direct edges have disjoint endpoint sets.

Final root counts satisfy $R \geq 2, B \geq 5$, and a non-direct port 2 occurs only at $R \geq 4$. Every non-direct port pair is $(0, 1)$ or $(0, 2)$ up to reversal. Its two allowed sets have size at least two, so every length at least three extends. At length two the midpoint is respectively $(1, 3)$ or $(2, 2)$ and the same inequalities verify both ends. This includes return paths at their legal lengths. Pendant port zero uses (0), $(0, 0)$, or opposite leaf port 2; pendant port one uses (1), $(1, 0)$, or leaf port zero, according as its length is one, two, or at least three. Direct checking or Lemma 10.1 verifies every case. □

12.3 A direct matching with ordinary paths of length four

Proof of the third statement. Let $h \in \{0, 1\}$ indicate a direct incidence, let l count return paths, and let z count all other root incidences. Thus $D = h + 2l + z$. Give direct edges weight 1, return ports $(0, 2)$, and choose the z free ports from $\{0, 2\}$. For a root with $h = 1$, if a free ports are 2 its counts are

$$(R, B) = (1 + 2l + 2a,\ 1 + 2l + 2(z - a)).$$

For a direct edge, if both ends have free ports and one has $z \geq 2$, fix the other end and try $a = 0, 1, 2$. At most two candidates are forbidden. If both have $z = 1$, their candidates are $(D - 1, D + 1)$ and $(D + 1, D - 1)$: use the same choice at different degrees and opposite choices at equal degrees. If

one end has $z = 0$, it has a return path. Change one such port pair from $(0, 2)$ to $(0, 1)$. That end now has even counts $(D - 1, D + 1)$ and the other keeps odd counts. If both have $z = 0$, change just one end. Thus every direct edge is legal, independently of all others.

At roots with $h = 0$, leave free ports zero; counts are $(2l, 2(l+z))$. The sole exceptional pendant is $l = z = 1$ with length two. Change its port to 1, making the root $(3, 3)$, and use $(1, 1)$. Every ordinary non-direct path has pure ports and length at least four, so the general path lemma applies even when its ports are equal. Unmodified returns have ports $(0, 2)$. A modified return has ports $(0, 1)$ at counts $(D - 1, D + 1)$; both allowed sets have at least two elements, so it also extends at length three. At a directly matched root, each colour used by a pendant pure port has odd degree at least three, different from the short pendant values two and four. At an unmatched root its blue degree is at least four, and equality to four on a two-edge pendant was exactly the exception repaired above. Long pendants use opposite pure leaf ports. These cases exhaust the construction. □

In all three proofs the case $W = \varnothing$ is separate: a single vertex has no edges; a two-edge path has word $(0, 0)$; a longer path uses opposite pure leaf ports. For a cycle, properly three-colour the position graph joining positions i and $i + 2$ cyclically. Its maximum degree is at most two and its resulting word satisfies every internal comparison. The excluded one-edge path is K_2.

13 Homogeneous direct cores with long connections

Lemma 13.1 (Even signed forms)**.** *Suppose each of finitely many homogeneous linear forms has an even number, at least four, of distinct variables, all coefficients are* ± 1*, and each variable occurs in at most two forms. There is a choice of signs* ± 1 *for the variables making every form nonzero.*

Proof. Make a signed multigraph whose vertices are forms and whose shared variables are edges; variables occurring once are boundary variables. In a component with a boundary variable, retain a spanning tree and one root boundary variable, set all other variables to 1, and process leaves upwards. The one remaining sign at each step avoids at most one forbidden value.

In a component without boundary variables every vertex has even degree at least four. Give each edge the product of its endpoint coefficients as sign. There is a positive cycle: if no fundamental cycle is positive, two negative fundamental cycles sharing a tree edge have a positive cycle as their symmetric difference; if none share tree edges the graph is a cactus and has a vertex of degree at most two, a contradiction. Keep such a cycle and a forest rooted on it. Eliminate off-cycle vertices as before. On the cycle the relations are $a_i y_i + b_i y_{i-1} + C_i \neq 0$ with even C_i. For $C_i = 0$ the allowed relation is a bijection on two signs; for $C_i = \pm 2$ it has three of the four pairs; for $|C_i| \geq 4$ all four pairs. If every relation is a bijection, positivity makes their composition have a fixed point. Otherwise their composition has at least three pairs: bijections preserve size, and composition of relations with at least three pairs has at least three pairs. Such a relation contains a diagonal pair. Hence the signs close around the cycle. □

Theorem 13.2 (A3)**.** *Let G have minimum degree at least two. In each component of $H = G[\{v : d_G(v) \geq 3\}]$, suppose the original degree is constant and the component is three-colourable. Suppose all ordinary non-direct connections have length at least three and there are no return paths. Then* $\operatorname{lir}(^2G) \leq 2$.

Proof. Orient each long connection and give it opposite pure ports. Its contribution $c(v)$ is twice an indegree. For an isolated direct-edge component uv of original degree D, the two long-incidence counts are both $D - 1$. Reversals of the long paths are sign variables, and $c(u) - c(v)$ is an even

form with $2(D-1) \geq 4$ incidences. If a long path joins u and v its sign occurs twice with combined coefficient ± 2; it is private to this equation. Temporarily omit equations with such variables. All remaining forms satisfy Lemma 13.1; then restore each omitted equation by its private variable, one of whose two signs makes it nonzero. Give each isolated direct edge weight 1. It is now legal in both colours.

For every larger direct component choose proper targets ϕ over $\mathbb{F}_3$ and solve the internal incidence equations $\sum_{e \ni v} x_e = \phi(v) - c(v)$. A nonbipartite component realises arbitrary demands: a spanning tree and an edge joining two vertices of the same tree parity allow adjustment of the signed total by an invertible coefficient ± 2, followed by leaf elimination. In a bipartite component solvability is precisely the alternating-sum condition in Lemma 7.1. Choose the larger side, of size at least two, initially all zero and the other side all one. Changing zero, one or two vertices of the first side to 2 changes that sum by $0, 2, 4$, covering all residues. The targets stay proper. Representatives $0, 1, 2$ of the solved edge weights give different red sums on every direct edge. The original degrees agree there, so the blue sums also differ. Isolated roots require no direct comparison. Finally fill all long paths with their opposite pure ports. □

For an r-regular simple mother graph, replacing selected edges by paths of length at least three is therefore allowed whenever the retained graph is three-colourable. At $r = 3$, a nonempty selected set leaves every retained component 2-degenerate, since a cubic retained component would be a whole connected component of the original graph. This recovers the earlier cubic subdivision constructions, already contained in A1. At unbounded degree the three-colourability hypothesis is substantive and has not been removed.

14 Balanced half-edges and simultaneous extensions

Lemma 14.1. *Let J be a multigraph, with loops allowed and no vertex of degree one. Its incidences admit values $0, 1, 2$ summing to $d_J(v)$ at every vertex, different at the two ends of every edge. Odd-degree vertices have exactly one incidence of value 1 and even-degree vertices none. For disjoint pairs of positive-degree vertices of the same parity, one may instead require exactly one vertex of each pair to have sum $d_J(v) \pm 1$, leaving every other sum unchanged.*

Proof. Balance-orient J, assigning 0 at tails and 2 at heads. At an odd vertex choose one majority-valued incidence to change to 1. There are $\lceil d/2 \rceil \geq 2$ different candidate edge identifiers; a loop cannot supply two majority-valued candidates. The request-edge bipartite graph has each request adjacent to at least two edges, each edge to at most two requests. Hall's condition follows by counting, so all changes can use distinct edges. The sums become exact and end values stay different.

For an odd-degree pair skip the correction at one member. For an even-degree pair create one request selecting any incidence at either member and changing it to 1; the union contains at least two edge identifiers. Each original vertex belongs to at most one request, so the same Hall argument applies. These changes give precisely the claimed one-unit deviations. □

Theorem 14.2 (A4)**.** *Let a connected multigraph M have original degrees $D(v) \geq 4$. Subdivide an edge set J into paths of length at least two, with all loops in J and loop paths of length at least three. Suppose J has no vertex of degree one, the retained graph $H = M - J$ is simple, and every retained edge has endpoints of different original degrees. Then the simple expanded graph has a legal doubled two-colouring with $R(v) = B(v) = D(v)$ at every old vertex.*

Proof. Use Lemma 14.1 on J and give every retained edge weight 1. Each old vertex receives $d_J + d_H = D$ in both colours. Retained edges compare different D values. Each subdivided path has distinct end weights and all four endpoint colour degrees at least four, so Lemma 10.1 fills it at every allowed length without changing the old sums. □

Theorem 14.3 (A5)**.** *Let H be a simple graph whose components of order at least three have given legal doubled two-colourings. On the same old vertex set let J be a connection multigraph with $k(v) = d_J(v)$ either zero or at least four, constant on each component of H. Require $k \geq 4$ on every K_2 component. Replace J edges by internally disjoint paths of length at least two, or at least three for loops. The colouring extends to the simple expanded graph preserving every given old edge weight. No initial colouring of K_2 is assumed.*

Proof. Pair the endpoints of the K_2 components in Lemma 14.1. All other old vertices receive exactly (k, k) from new ports. Every given old colour-degree difference is unchanged, since k is equal at its two ends. Give each old K_2 edge weight 1. Its endpoints then have counts $(k+1+\epsilon, k+1-\epsilon)$ and $(k+1, k+1)$ for $\epsilon \in \{-1, 1\}$, so that edge is legal. All vertices incident with new paths have both colour degrees at least four. Different new end ports can therefore be filled at length two by Lemma 10.1. All old and new checks hold together. □

A vertex-minimal counterexample cannot admit the latter decomposition with $J \neq \varnothing$: deleting its new internal vertices gives smaller old components, while the K_2 and isolated cases are already handled. This does not assert that every graph has such a decomposition.

15 Near-centre parity assignments

Lemma 15.1. *A parity prescription $p : V(G) \to \{0, 1\}$ can be realised by weights $0, 1, 2$ with $R(v) \equiv B(v) \equiv p(v) \pmod 2$ and $|R(v) - d_G(v)| \leq 1$ if and only if $\sum_{v \in C} p(v)$ is even in every component C. The edges changed from weight 1 may be restricted to a forest.*

Proof. Necessity is the red handshake identity. Start with all weights 1. In each spanning tree choose a T-join for $T = \{v : p(v) \neq d_G(v) \pmod 2\}$ by leaf elimination; $|T|$ is even. Pair incidences of the chosen forest at each vertex, leaving one unpaired at each odd-degree vertex. Its edges decompose into open trails with those unpaired ends; there is no closed trail in a forest. Along each trail change weights alternately to 2 and zero. Contributions cancel at every paired passage, leaving a change of ± 1 exactly at the vertices of T. All claims follow, including the exact centred assignment when T is empty. □

Theorem 15.2 (A6)**.** *Let F consist of the original edges uv with $|d_G(u) - d_G(v)| \leq 2$. If F is bipartite and its bipartition phases can be chosen to give an even total number of odd labels, then ${}^2G$ has a legal two-colouring. Both colours distinguish the ends of every original edge.*

Proof. Realise those labels by Lemma 15.1. On F, both colour degrees have different parities. Off F, the intervals $[d_G(v) - 1, d_G(v) + 1]$ at the two ends are disjoint, and contain both colour degrees. This verifies every comparison. □

The phase condition is exact: an odd-order component of F allows the total parity to be changed by a flip. If all components have even order, it holds exactly when an even number of components have two odd-sized sides. For example, connect the roots of three disjoint K_{r_i, r_i} modules in a triangle, where the $r_i \geq 4$ are even and pairwise differ by at least four. The graph has no degree-two threads and is nonbipartite. Its F consists precisely of the three modules with even sides, so the theorem applies. This condition does not cover general nonbipartite regular cores.

16 Clique interfaces, block trees and capacity

Fix a root z of K_n, external contributions $A, B \geq 0$ with $A + B$ even, and a desired internal root red sum S. Only internal edges are to be checked, using final root counts $(A+S, B+2(n-1)-S)$.

Theorem 16.1 (Complete clique root interface)**.** *For every $n \geq 4$ and $0 \leq S \leq 2(n-1)$ there is a legal internal colouring with the prescribed root sum. For $n \geq 5$ every nonroot vertex may additionally have both colour degrees in $[1, 2n-3]$. For $n = 4$ the same strengthening holds except at $(A, B, S) = (1, 3, 4), (3, 1, 2)$; the ordinary interface still holds there.*

Proof. For $n = 4, 5$, put $r = n-1$. External counts exceeding $2r$ are comparison-equivalent to $2r+1$ for odd counts and $2r+2$ for even counts, since every nonroot count is at most $2r$. Consequently the entire base domain is

$$A, B \in \{0, \ldots, 2r+2\}, \quad A + B \equiv 0 \pmod 2, \quad S \in \{0, \ldots, 2r\}.$$

It contains $287 + 549 = 836$ entries. Actual edge-weight witnesses are in `rooted_clique_interfaces.json`. The independent standard-library checker `rooted_clique_certificate.py` verifies equality with this whole domain, root sums, every actual colour comparison, and every nonroot bound outside the two exceptions. Thus the bases include arbitrarily large physical external counts by exact clipping, not by an unproved extrapolation.

For $n \geq 6$ use strong induction by two vertices. If $2 \leq S \leq 2n-4$, take a rooted K_{n-2} with the same external counts and internal sum $S-2$. Add u, v, set $wu = 0, wv = 2$ for every old w, and $uv = 1$. All old colour degrees increase by two; the new counts are

$$u : (1, 2n-3), \qquad v : (2n-3, 1).$$

Old nonroot counts are within $[2, 2n-4]$, so all their new edges are legal. Write the final root counts as p, q. If neither equals $2n-3$, its two new pure edges are also legal. Otherwise both p, q are odd because $p + q$ is even. Choose an old nonroot w_0 and change its two new weights from $(0, 2)$ to $(1, 1)$. All old counts stay fixed, while the new counts become

$$u : (2, 2n-4), \qquad v : (2n-4, 2).$$

Inductively strengthened old nonroot counts, after adding two, lie in $[3, 2n-5]$ and are strictly separated from these boundary values. The root counts are odd, so its comparisons with the new even values hold. When $n = 6$ and the old K_4 is one of the two exceptional bases, the final root counts are instead $p = q = 7$, different from $2n-3 = 9$; the unmodified construction already works. Thus no missing strengthened base is needed.

For $S = 0, 1$, take the old K_{n-2} with the same internal sum but formal external counts $A' = A - 2, B' = B + 2$. If $A \geq 2$, these are physical. If $A = 0, 1$, obtain the old witness using the nonnegative replacement $\widehat{A} = A \bmod 2$ and then reinterpret its root red count with A'. This keeps the old internal edges legal: for $S = 0$ the root has no red edge; for $S = 1$ its sole red incidence meets a vertex of red degree at least one, whereas the reinterpreted root red count is nonpositive. Its blue count is unchanged. These formal negative offsets are only an intermediate comparison argument.

Add u, v with $wu = 0, wv = 2$ for old nonroots, $zu = zv = 0$, and $uv = 2$. Relative to the reinterpreted old counts every old vertex gains two in each colour, so the root now has exactly the originally specified physical counts and internal sum S. The new counts are $(2, 2n-4), (2n-4, 2)$. The old nonroot counts lie in $[3, 2n-5]$; the root blue count is at least $2n-3$. All new comparisons follow. The old base parameter $S = 0$ or 1 is never an exceptional K_4 parameter. Finally obtain $S = 2n-3, 2n-2$ by exchanging the colours and A, B. These cases cover every S and preserve the asserted nonroot bounds. □

16.1 Bouquets and arbitrary cores

Theorem 16.2 (A8)**.** *At a vertex of old degree h in a given legal colouring, attach cliques of orders $n_i \geq 4$ sharing only that vertex, with total new degree $L = \sum_i (n_i - 1) \geq h$. Every old edge colour can be preserved. Alternatively, for an arbitrary uncoloured simple core C, attach such a bouquet at every vertex v with $L_v \geq d_C(v)$. The entire expanded graph can then be coloured.*

Proof. For the first assertion, let the old counts at the root be a, b. An added red contribution $s \in [0, 2L]$ gives final counts $(a + s, b + 2L - s)$. Each old pure edge forbids at most one value of s and each mixed edge at most two. There are at most $2h$ forbidden values and $2L + 1 > 2h$ candidates. Choose an allowed s and split it into $s_i \in [0, 2(n_i - 1)]$. Clique i receives external counts

$$A_i = a + s - s_i, \quad B_i = b + 2L - s - \big(2(n_i - 1) - s_i\big).$$

These are nonnegative with even sum. Theorem 16.1 colours it with internal sum s_i. Every clique uses the same final root counts and all old neighbours keep their counts.

For the second assertion give every core edge weight 1. Put $h_v = d_C(v)$ and $D_v = h_v + L_v$. In any vertex order choose $R(v) \in [h_v, h_v + 2L_v]$. A previously processed neighbour forbids at most $R(w)$ and $2D_v - B(w)$, so at most $2h_v$ values are forbidden. There are at least $2h_v + 1$ candidates. Choose all red targets greedily and set $B(v) = 2D_v - R(v)$. Every mixed core edge now distinguishes both colours. Split the remaining red contribution among the cliques and realise it by the root interface theorem. □

The first capacity bound cannot uniformly be reduced to $L = h - 1$ for $h \geq 4$. Pair the integers $0, \ldots, 2h - 2$ within their even and odd subsequences using adjacent values, covering a final odd triple by two overlapping pairs. This gives exactly h pairs (a_i, b_i) with $a_i = b_i + 2$, distinct a_i, distinct b_i, and union $\{0, \ldots, 2h - 2\}$. On a central K_{h+1} with vertices $v, u_1, \ldots, u_h$ put every weight 1. Attach a rooted K_{h+1} at each u_i, with internal red sum a_i using the theorem. The old counts are

$$v : (h, h), \qquad u_i : (h + a_i,\ 3h - 2 - b_i).$$

All central mixed edges are legal: both lists at the u_i are distinct and at least $h + 2$. Add K_h at v. Its contribution $s \in [0, 2h - 2]$ is forbidden by vu_i precisely at $s = a_i$ or $s = b_i$. The pairs cover every candidate. Nonetheless the entire graph is colourable by the block-tree theorem below, rooted at v: each nonroot cutvertex has equal parent and child capacities. Thus this is a sharp preserved-colour obstruction, not a graph obstruction.

16.2 Clique block trees with full root freedom

Theorem 16.3 (A7)**.** *Let every block of a connected graph be a clique of order at least four. Root its block-vertex incidence tree at any vertex z. At every nonroot cutvertex require the sum of its child-block degrees $\sum(n_i - 1)$ to be at least its parent-block degree $n_0 - 1$. For all physical external counts A, B at z and all $S \in [0, 2d_G(z)]$, the graph has a legal internal colouring with root internal red sum S. In particular all equal-order clique block trees satisfy this conclusion.*

Proof. First colour the bouquet of root blocks, distributing the specified sum S between them and using the root interface with the contributions of the other root blocks included as external counts. Process the other cutvertices outwards. Before its children are added a cutvertex belongs to its one parent block and has current degree $h = n_0 - 1$. The child bouquet has capacity at least h, so Theorem 16.2 extends the current colouring without changing any old edge weight. Actual external

root counts remain included in all old comparisons. Adding blocks away from z does not change its internal red sum. The finite incidence tree is eventually exhausted. □

In a vertex-minimal counterexample, any bouquet of leaf clique blocks of order at least four at a vertex must have capacity $L < h$, where h is its degree in the remainder. Otherwise delete the leaf blocks, colour the smaller remainder, and apply Theorem 16.2. If the remainder is K_1, use the root interface directly. If it is K_2, give its edge weight 1 and the bouquet total red contribution 1; the endpoints have counts $(2, 2L)$ and $(1, 1)$, so the old edge is legal and all cliques can be filled. Hence no hidden K_2 assumption is used in this minimality argument.

17 Gluing coloured pieces along a tree

Let connected pieces P_i partition the edges of G. Let U be the vertices belonging to more than one piece; all other vertices are called private. Suppose the bipartite incidence graph of pieces and shared vertices is a tree. In the given local weighting of P_i, write d_i, r_i, b_i for the local degrees and set

$$a_i(v) = r_i(v) - d_i(v), \quad M_v = \max_{i:v\in P_i} |a_i(v)|, \quad I_v = [d_G(v) - M_v, d_G(v) + M_v] \quad (v \in U).$$

Theorem 17.1 (A9)**.** *One may obtain a legal colouring solely by whole-piece colour flips provided that: private–private edges are locally legal; each edge from shared v to private u in P_i satisfies $d_G(v) - M_v > \max(r_i(u), b_i(u))$; and each shared–shared edge uv satisfies $|d_G(u) - d_G(v)| > M_u + M_v$.*

Proof. Root the incidence tree at a piece. On reaching a shared vertex its parent piece's flip is already fixed. Starting with its contribution $s = \pm a_i(v)$, choose the flip of each child piece to minimise the new absolute sum. The identity

$$\min\{|s + a|, |s - a|\} \le \max\{|s|, |a|\}$$

keeps $|s| \le M_v$. Each child piece has only one parent shared vertex, so no choice is made twice. At the end the global red and blue degrees at v are $d_G(v) \pm s$, both in I_v. Private counts are their local counts up to a swap. The three edge types in the hypotheses are respectively preserved by local legality, strict separation above private counts, or disjoint shared intervals. Every edge is therefore legal. □

If the given pieces have maximum degree at most q, a sufficient uniform condition is $d_G(v) \ge 3q + 1$ for shared vertices and degree gap at least $2q + 1$ on shared–shared edges. Indeed $M_v \le q$ and private colour degrees are at most $2q$. Taking known colourings of cycle or cubic blocks gives, for example, the conditions shared degree at least 10 and adjacent shared-degree gap at least 7. This allows unbounded global maximum degree and pieces that are not cliques.

17.1 Why whole-piece flips do not suffice in general

For each $r \ge 2$, identify the distinguished left vertex z of two copies of $K_{r,r}$. In each copy put weight 2 in column 1, weight zero in column 2, weight 1 in all remaining entries of row z, and zero elsewhere. The local red sums are r at z, 2 at other rows, $2r$ at column 1, zero at column 2, and 1 at other columns. As the local original degrees are equal, these values verify local legality on every edge.

The local root counts are (r, r) even after a whole-piece flip. After identification the root counts are $(2r, 2r)$, and in each flipped or unflipped copy it still has a same-colour neighbour of that colour

degree $2r$. Thus all four choices of whole-piece flips fail, although the incidence graph is a tree. For $r \geq 3$, changing one entry of row z in a remaining column from 1 to zero in one copy repairs the whole graph: the root becomes $(2r-1, 2r+1)$, the changed column's red sum becomes zero, and all other local counts persist. At $r = 2$ replace one copy's matrix by rows $(0,0), (2,2)$. These direct repairs show why the obstruction concerns given-piece gluing and not the conjecture.

18 Fixed targets, factors and exchanges

This section gives applications of classical matching and factor theory. These applications distinguish three tasks: independently selecting mutually distinguishing targets, realising one target vector by shared edge weights, and finding a target for which the actual colour constraints hold. Only the last task for every graph would prove the general conjecture.

18.1 Independent target lists are not edge realisations

Lemma 18.1. *Give each vertex v of a simple graph H a real number D_v and a list L_v of at least $d_H(v)+1$ distinct real numbers. One can choose $z_v \in L_v$ such that both z_v and $D_v - z_v$ distinguish the endpoints of every edge.*

Proof. Orient the edges arbitrarily and write $Q(z) = \prod_{uv}(z_u - z_v) = \sum_\alpha q_\alpha z^\alpha$. For $d = (d_H(v))_v$ and $m = |E(H)|$, $z^d Q(z^{-1}) = (-1)^m Q(z)$, so $q_{d-\alpha} = (-1)^m q_\alpha$. The coefficient of z^d in the highest homogeneous part of $Q(z)Q(D-z)$ is therefore $\sum_\alpha q_\alpha^2 > 0$. Polynomial interpolation on the lists, equivalently the Combinatorial Nullstellensatz [7], gives a nonzero evaluation. □

This lemma says nothing about incidence realisation. On a triangle, the lists $\{0,2,4\}$ with $D_v = 4$ allow mutually distinct complementary targets, but the targets $0,2,4$ cannot be realised by weights $0,1,2$: the zero vertex forces its two weights zero, preventing its degree-four neighbour from receiving four.

18.2 A precise fixed-target criterion

Consider a simple subgraph H whose vertices receive fixed external counts $a(v), b(v)$. Specify total red targets $R(v)$ and put

$$q(v) = R(v) - a(v), \qquad B(v) = b(v) + 2d_H(v) - q(v).$$

Define $E_0 = \{uv : R(u) = R(v)\}$ and $E_2 = \{uv : B(u) = B(v)\}$. If the two sets overlap, no legal realisation of these targets exists. Otherwise set

$$K = H - (E_0 \cup E_2), \qquad t(v) = q(v) - 2d_{E_2}(v).$$

Proposition 18.2. *The targets admit a legal internal colouring exactly when $q(v) \in [0, 2d_H(v)] \cap \mathbb{Z}$, $E_0 \cap E_2 = \varnothing$, and ${}^2K$ has a spanning subgraph of degrees $t(v)$.*

Proof. On E_0 all weights are forced to zero, on E_2 to 2. Every remaining edge distinguishes both total colour degrees and is free to have any weight. Its selected red copies must supply exactly t. This correspondence is reversible. □

Here is an explicit ordinary matching reduction. For each copy of ${}^2K$ create its two endpoint ports and join them. At v create $2d_K(v) - t(v)$ auxiliary vertices adjacent to all ports at v. A

perfect matching uses that many local ports, leaving exactly $t(v)$ ports matched across original copies. Select those copies for the factor. Conversely any such factor completes to a perfect matching by matching unused ports locally. The auxiliary graph has at most $8|E(K)|$ vertices. Range failures for t are rejected first.

18.3 Capacity and odd-component certificates

For disjoint $S, T \subseteq V(K)$ put

$$g_t(S,T) = t(S) + 2\sum_{v\in T} d_{K-S}(v) - t(T),$$

and let $o_t(S,T)$ be the number of components C of $K - S - T$ for which $t(C)$ is odd.

Proposition 18.3. *Real weights $0 \le x_e \le 2$ realising t exist if and only if $g_t(S,T) \ge 0$ for all disjoint S,T. Integer weights exist if and only if $g_t(S,T) \ge o_t(S,T)$ for all such pairs.*

Proof. Necessity of capacity follows by subtracting the sums over T and S and bounding the remaining terms by $4|E_K(T)| + 2|E_K(T, V \setminus (S \cup T))|$. For sufficiency, let $\mathcal{Z} = \{Bx : x \in [0,2]^{E(K)}\}$, where B is the unsigned incidence matrix. Its support function at y is $2\sum_{uv} \max(0, y_u + y_v)$. If $t \notin \mathcal{Z}$, a separating vector y violates this bound. Write y as the integral, over $h \ge 0$, of the vector taking value 1 where $y_v > h$, value -1 where $y_v < -h$, and zero elsewhere. For each edge the same integral identity holds for the positive part of $y_u + y_v$; it is checked separately when the two signs agree or disagree. Some three-valued vector must therefore violate the support bound, which is precisely a violated $g_t(S,T) \ge 0$.

For integers apply Tutte's factor theorem [6] to ${}^2K$. In its odd-component term the number of edges from a component to T is even because every original edge is doubled. The odd term is consequently exactly $o_t(S,T)$, and its degree term is g_t. This proves the stated equivalence. □

For even total target sum, $g_t - o_t$ is even. Colour complementation sends t to $2d_K - t$ and interchanges S, T, preserving both quantities. If every $t(v)$ is even, every odd-component term is zero, so continuous feasibility already gives an integer realisation. One can also see this directly from a vertex of the capacity polytope. Strictly fractional edges have independent incidence columns, hence each of their components is a tree or has one odd cycle. Leaves are impossible with even residual demands and all other weights at 0 or 2. The fractional support is thus a disjoint union of odd cycles; each adjacent pair sums to 2, forcing all their weights to 1. The resulting extreme point is integer. This is a fixed even-target integrality fact, not permission to require all red degrees even in the original conjecture.

Prescribed parity intervals also reduce to matching. Tighten $[\ell(v), u(v)] \cap [0, 2d(v)]$ to its first and last values $g(v), f(v)$ of the desired parity. Add $(f(v) - g(v))/2$ virtual loops at v to ${}^2G$. An exact factor of degree f becomes the required interval assignment after virtual loops are deleted, and every such assignment can be completed by loops. For a matching gadget with loops, create two ports per edge or loop, join each port pair, and create $f(v)$ demand vertices joined to all local ports. Ports are either matched together (edge unselected) or both to demands (selected); exactly $f(v)$ incidences are selected. This polynomial reduction includes degree feasibility only.

18.4 Minimum conflicts and fixed-degree exchanges

Let $\mathcal{D}(G)$ be all realisable integer red target vectors, and

$$\Psi(x) = \sum_{uv} \left(x_{uv} \nVdash_{R(u)=R(v)} + (2 - x_{uv}) \nVdash_{B(u)=B(v)} \right), \qquad F(q) = \min_{Bx=q} \Psi(x).$$

For a fixed q put $I_R(uv) = \mathbb{1}_{q(u)=q(v)}$, $I_B(uv) = \mathbb{1}_{2d(u)-q(u)=2d(v)-q(v)}$. Then

$$F(q) = 2\sum_e I_B(e) + \min_{Bx=q} \sum_e (I_R(e) - I_B(e))x_e. \tag{22}$$

The matching gadget above, with weight $-(I_R - I_B)$ on each cross-port pair and zero on local edges, computes the minimum by a maximum-weight perfect matching. This is an exact application of weighted matching; the costs change when q changes.

If two assignments realise the same target, mark their differing copies by $+$ or $-$ and pair opposite signs at each vertex. They decompose into even alternating closed walks. Altering one walk moves toward the other assignment, never leaves $\{0,1,2\}$, and preserves all degrees. Repeated vertices and repeated original edges must be allowed. At fixed targets a legal realisation, if it exists, can be reached without increasing Ψ, since all forced edges move monotonically to their required extremes and all free edges permit both colours.

More generally, for $q, r \in \mathcal{D}(G)$ and any unit step s toward r, there is a second unit step t toward r such that

$$q + s + t,\ r - s - t \in \mathcal{D}(G), \qquad \|q + s + t - r\|_1 = \|q - r\|_1 - 2.$$

To prove this, pair opposite marked incidences as far as possible. Unpaired incidences have exactly the signs of $r(v) - q(v)$. Start at one representing s, follow its alternating paired trail, and stop at the next unpaired incidence, representing t. Transfer the marked copies in opposite directions in the two assignments. Every edge stays between its two initial values. This is the standard constant parity jump-system exchange phenomenon; see [8]. In particular the two-unit target graph is connected, with distance $\|q - r\|_1/2$. For a *fixed* linear edge-cost vector c the same transfer gives

$$f_c(q + s + t) + f_c(r - s - t) \le f_c(q) + f_c(r), \quad f_c(q) = \min_{Bx=q} c \cdot x,$$

because the two assignments have conserved total cost. The function F in (22) does not share this conclusion in general; Section 27 gives explicit failures.

18.5 A parity-defect lower bound

For $E_0 \cap E_2 = \varnothing$, contract every component of the free graph K. Let $\mathcal{O}$ be the resulting vertices with odd q sum. On a forced edge write $z_e = x_e$ for $e \in E_0$ and $z_e = 2 - x_e$ for $e \in E_2$. Then $\Psi = \sum z_e$, and each free component C satisfies

$$q(C) \equiv \sum_{e \in \delta(C)} z_e \pmod 2.$$

After duplicate copies cancel modulo two, the bad forced copies contain an $\mathcal{O}$-join in the quotient graph. Therefore $F(q)$ is at least its minimum join cardinality. Pairing incidences decomposes any join into paths between paired odd vertices and closed parts; conversely shortest paths of a minimum perfect matching on the odd vertices, cancelled modulo two, give a join. This proves the familiar shortest-path matching formula for this lower bound. It detects parity only; internal factor capacities can require a larger defect.

19 Normal forms and conditional structural reductions

19.1 Sparse mixed-edge normal form

Proposition 19.1. *Every assignment can be changed, preserving all colour degrees and not increasing Ψ, until the weight-one edges form a cactus whose cycles are vertex-disjoint odd cycles.*

Proof. On an even closed trail of mixed edges alternately increase and decrease weights by one. Both signs of this change are feasible and preserve degrees. At fixed degrees Ψ is linear, so one choice does not increase it. Every traversed edge ceases to be mixed. Repeat until no even closed trail remains. There is no even cycle. Two cycles sharing at least two vertices give a theta subgraph, two of whose three paths have the same parity, giving an even cycle. Two odd cycles sharing exactly one vertex give an even closed trail. Thus cycles are vertex-disjoint and odd, and all other edges of each mixed component are bridges. □

If there are n vertices, c mixed components including isolates, and k mixed cycles, their edge count is $n - c + k \le n - c + \lfloor n/3 \rfloor$. This sparsification neither asserts nor needs $\Psi = 0$.

19.2 Energy cleanup and an exact one-edge identity

Define $\mathcal{E}(x) = \sum_v (R(v) - d_G(v))^2$. If a present colour has equal degree $k > \max(d(u), d(v))$ at both ends of uv, changing one copy of that colour to the other changes the energy by

$$-4k + 2(d(u) + d(v)) + 2 \le -2.$$

If a present colour has equal degree one, its edge is mixed. Changing the other copy to that colour changes the energy by $6 - 2(d(u) + d(v)) \le 0$, provided the graph has no isolated K_2 component. Both new endpoint colour degrees are even, so no new degree-one conflict appears. Hence the lexicographic pair $(\mathcal{E}, h_1)$, where h_1 counts bad degree-one copies, strictly decreases under these two rules. Termination takes at most $(\lfloor \mathcal{E}(x)/2 \rfloor + 1)(2m + 1) - 1$ steps. The remaining bad colour degrees are between two and the larger endpoint original degree. The mixed-edge normal form can subsequently be imposed, preserving these statements because it fixes all degrees and changes only edges that already had both colours.

The energy is not Ψ. On a four-vertex path, changing the word $(0, 1, 0)$ to $(0, 2, 0)$ lowers $\mathcal{E}$ from two to zero while increasing Ψ from two to six. The all-mixed triangle has energy zero and six bad copies and is already stopped by both rules.

For completeness there is an exact local accounting identity. For a permitted change $x_e \mapsto x_e + \eta$, $e = uv$, $\eta = \pm 1$, put

$$H_z^\eta(e) = \sum_{zw \ne e} \big[x_{zw} (\mathbb{1}_{R(w)=R(z)+\eta} - \mathbb{1}_{R(w)=R(z)}) + (2 - x_{zw})(\mathbb{1}_{B(w)=B(z)-\eta} - \mathbb{1}_{B(w)=B(z)}) \big].$$

Then

$$\Psi(x') - \Psi(x) = \eta(\mathbb{1}_{R(u)=R(v)} - \mathbb{1}_{B(u)=B(v)}) + H_u^\eta(e) + H_v^\eta(e).$$

On e the degree differences stay fixed and only multiplicities change; the remaining affected edges are exactly those in the two sums. Any claimed improvement must account for these neighbour terms.

19.3 Exact elimination of peripheral trees

If G contains a cycle, its 2-core C is nonempty and connected. Every component outside it is a tree attached by exactly one edge: a cycle outside or two attachments would enlarge the minimum-degree-two subgraph. At a core vertex put $h = d_C(v) \ge 2$ and let ℓ, s, b count attached trees of

orders one, two and at least three, respectively. Let $k = \ell + s + b, D = h + k$. At a proposed final red degree q put $B = 2D - q$ and define

$$L_1(q) = \{c \in \{0,1,2\} : c = 0 \Rightarrow B \neq 2,\ c = 1 \Rightarrow q, B \neq 1,\ c = 2 \Rightarrow q \neq 2\},$$
$$L_2(q) = \{c \in \{0,1,2\} : c = 0 \Rightarrow B \neq 4,\ c = 2 \Rightarrow q \neq 4\},$$
$$S_v(q) = \ell L_1(q) + sL_2(q) + b\{0,1,2\}, \qquad J_v(q) = S_v(q) \cap [q - 2h, q].$$

The sums are Minkowski sums, with $0L = \{0\}$. A graph colouring is equivalent to core weights y_e and targets q_v satisfying

$$q_v - \sum_{e \in C, e \ni v} y_e \in J_v(q_v)$$

and the actual red/blue comparison on every core edge with total degree D_v. Necessity is restriction. For sufficiency split the chosen sum into allowed bridge weights and apply the tree interface lemma independently. Physical counts hold because $q = t + \sum c_i \geq c_i$ and $B = (2h - t) + \sum(2 - c_i) \geq 2 - c_i$.

Each $S_v(q)$ is an integer interval or an interval of step two. Indeed L_1 is an interval or $\{0,2\}$, and L_2 is an interval, possibly $\{1\}$. A width of at least one in any other interval fills the gaps of a sum of $\{0,2\}$ sets. This is an exact compact description, not an approximation of feasible root degrees. Trees of order at least three may therefore be replaced by any rooted three-vertex path, preserving colourability in both directions.

19.4 A sharp tree capacity extension

Theorem 19.2. *At a core vertex of core degree $h \geq 2$, $k \geq h$ attached rooted trees can be restored to a given legal colouring of the graph with those trees removed, preserving every old edge colour.*

Proof. Fix its old internal red degree $t \in [0, 2h]$. The feasible final targets are $\mathcal{Q}(t) = \{q \in [t, t+2k] : q - t \in J_v(q)\}$. For $k \geq 3$ this is the whole interval, except for: value 1 if all branches are singletons and $t = 0$; its colour-complement at $t = 2h$; value 4 if there are exactly three branches, all of order two, and $t = 0$; its complement at $t = 2h$. To check this list only the low targets $1, 2, 4$ and their complements need inspection. At $q = 1$ only singleton bridge sets become $\{0,2\}$, causing a gap exactly when all branches are singletons and $t = 0$. At $q = 2$ their total upper bound is $2k - \ell \geq k \geq 3$, sufficient for any $2 - t$. At $q = 4$ the upper bound is $2k - s$, insufficient only when $k = s = 3, t = 0$. All other targets allow all weights. Low and high cases are disjoint since $h + k \geq 5$.

If j old incident edges are mixed, old neighbours forbid at most $h + j$ new targets. When $0 < t < 2h$, there are $2k + 1 > h + j$ candidates. At $t = 0$ or $2h$ no old edge is mixed and at least $2k > h$ remain. The small case $h = k = 2$ has at least four candidates, and all five at $t = 2$: directly inspect $q = 1, 2, 4, 6, 7$ in the bridge sets. At $t = 1, 3$ exactly one old edge is mixed, giving at most three forbidden targets; at $t = 0, 4$ there are at most two. Thus an allowed target always exists. Split its bridge contribution and fill the trees. □

The bound cannot uniformly be reduced to $k = h - 1$ for any $h \geq 2$. Create roots $v, u_1, \ldots, u_h, z$, edges vu_i, and, for each i, $d_i - 1$ internally disjoint two-edge paths from u_i to z, where $d_i = 2h - 2 + i$. The old degrees are $h, d_i, 5h(h-1)/2$ at v, u_i, z and two at the path interiors. All-mixed old edges are legal. Attach $h - 1$ three-vertex rooted paths at v. A preserved old colouring has $q \in [h, 3h - 2]$ and total colour sum $4h - 2$ there. The old edges forbid $q = d_i$ and $q = 4h - 2 - d_i$, covering that entire interval. Yet a full recolouring exists: change only vu_2 to zero, give new bridges weight 1 and each new internal path word $(0,0)$. Now v has $(2h - 2, 2h)$, u_2 has $(2h - 1, 2h + 1)$, other u_i keep (d_i, d_i), and each tree has successive counts $(1,3), (0,4), (0,2)$. These values verify every edge.

19.5 Universal vertices and minimality

Theorem 19.3. *If z is universal in $G \neq K_2$, and every component of $G - z$ of order at least three has a given legal colouring, then the colouring extends to G, preserving those given edge weights. Isolated vertices and K_2 components require no given colouring.*

Proof. Let $H = G - z$ have N vertices, k components K_2, ℓ isolates and larger components of orders c_i. Join every larger component and isolate to z with blue edges. For each $K_2 = uv$ put $uv = 1, zu = 0, zv = 2$. Its endpoints become $(1,3),(3,1)$ and contribute $(2,2)$ to z. Thus z has $(2k, 2(N-k))$. The K_2 interfaces compare even counts to three; an isolated blue neighbour has count two, equal only in the excluded case $G = K_2$. In a larger component, all old blue degrees increase by two, preserving every old comparison, and are at most $2c_i$. If H has any other component, $2(N-k) > 2c_i$, so the new interfaces are legal.

It remains that H is a single connected component of order $N \geq 3$. All-blue new edges fail only if an old vertex is universal in H and has every old copy blue. If no such vertex exists, retain them. Otherwise make every new edge red. This preserves old differences and can fail only at an old all-red universal vertex. An all-blue and an all-red universal vertex cannot coexist, since their common positive-multiplicity edge would need both pure colours. Hence the second choice works. □

Consequently a vertex-minimal counterexample has no universal vertex, so $\Delta \leq n-2$. It has a nonempty 2-core, every attached tree has at most three vertices, and the number k_v of trees at a core vertex is smaller than its core degree h_v. These follow by replacement and Theorem 19.2; the smaller remainders contain a cycle and are not K_2. Hence, for core order and size n_C, m_C,

$$|V(G)| \leq n_C + 3\sum_v (h_v - 1) = 6m_C - 2n_C.$$

Every maximal degree-two core thread has length at most five, including return threads. At length at least six replace its interior and attached trees by a new bare two-edge path, or a triangle for a return. The smaller graph is simple and connected and has a legal colouring by minimality. Its fixed endpoint counts and end weights extend back by Theorem 11.2. The condition $k_v < h_v$ ensures that each internal core-degree-two vertex has at most one tree. If the core is a cycle, its length is also at most five. For a noncycle core compressed to a skeleton with n_H vertices, m_H edges and minimum degree at least three, there are at most $4m_H$ degree-two vertices. Combining the branch bound gives

$$|V(G)| \leq 22m_H - 2n_H.$$

The skeleton size and degrees remain unbounded; none of these necessary conditions resolves an arbitrary minimum-degree-three core.

20 Exact polynomial certificates and parity identities

Orient each original edge $e = uv$ arbitrarily and put $\Delta_e = d_G(u) - d_G(v)$ and $D_e = R(u) - R(v)$. The linear form D_e does not involve x_e, whose contributions cancel.

Proposition 20.1. *The quadratic factor*

$$f_e = D_e^2 - 2\Delta_e(2 - x_e)D_e + 2\Delta_e^2(x_e - 1)(x_e - 2)$$

is nonzero exactly when the actual colour condition holds on e, for $x_e \in \{0,1,2\}$. Consequently $\prod_e f_e$ has a nonzero point on that grid if and only if ^2G has a legal two-colouring.

Proof. Its evaluations at $x_e = 0, 1, 2$ are respectively $(D_e - 2\Delta_e)^2$, $D_e(D_e - 2\Delta_e)$, and D_e^2. These express exactly the blue-only, mixed and red-only comparisons. □

For arbitrary real parameters τ_e define

$$f_{e,\tau} = D_e(D_e - 2\Delta_e) + \tau_e(x_e - 1)(D_e + \Delta_e(x_e - 2)),$$

and the multilinear parameter polynomial

$$C_G(\tau) = \left[\prod_e x_e^2\right] \prod_e (D_e^2 + \tau_e x_e D_e + \tau_e \Delta_e x_e^2). \tag{23}$$

The three evaluations of $f_{e,\tau}$ are $(D_e - 2\Delta_e)(D_e - \tau_e)$, $D_e(D_e - 2\Delta_e)$, and $D_e(D_e + \tau_e - 2\Delta_e)$. Nonzero therefore always implies actual legality, although additional legal assignments may be excluded.

Proposition 20.2. *If C_G is not the zero polynomial, $\operatorname{lir}(^2G) \le 2$. Its nonvanishing is equivalent to nonvanishing at some $\tau \in \{0,1\}^{E(G)}$. The fixed coefficient $c_G = C_G(2\Delta)$ is a sufficient special case.*

Proof. For every polynomial of total degree at most $2m$,

$$\left[\prod_e x_e^2\right] P = \sum_{a \in \{0,1,2\}^{E(G)}} \frac{P(a)}{\prod_e w(a_e)}, \qquad w(0) = 2,\ w(1) = -1,\ w(2) = 2. \tag{24}$$

In one variable the weighted sums of powers of degrees zero and one vanish and that of degree two equals one. A total-degree-at-most-$2m$ monomial with any exponent above two has another below two, so only the all-two monomial contributes. This proves the identity. Apply it to $\prod f_{e,\tau}$: its leading coefficient is (23). A nonzero coefficient forces a nonzero grid value. The assertion about binary parameters follows from unique interpolation of multilinear polynomials. This is an application of [7]; no general nonzero-coefficient theorem is supplied here. □

Write $D = Mx$. On the binary-copy variables $z_{e,1}, z_{e,2}$ put $x_e = z_{e,1} + z_{e,2}$. The copy comparison is the nonzero linear expression $D_e - 2\Delta_e + 2\Delta_e z_{e,i}$. Its coefficient matrix is

$$\mathcal{A}_G = M \otimes J_2 + 2\,\mathrm{diag}(\Delta) \otimes I_2.$$

The coefficient of the full squarefree monomial is its permanent, and

$$\operatorname{per}(\mathcal{A}_G) = 2^m c_G. \tag{25}$$

To check this equality, the quadratic part of $f_e(a+b)$ differs from the product of the two copy linear parts only by $2\Delta_e^2(a^2 + b^2)$, which cannot contribute to a squarefree monomial. Each substitution $x_e^2 = (a+b)^2$ contributes a factor two to the desired coefficient. Reversing an edge reverses its two copy rows, leaving the permanent unchanged.

20.1 A principal-minor formula modulo two

Let A and L be the adjacency and Laplacian matrices of the line graph of G, indexed by original edges; all determinants in the next statement are over $\mathbb{F}_2$.

Theorem 20.3. *For every $S \subseteq E(G)$,*

$$[\prod_{e\in S} \tau_e] C_G(\tau) \equiv \det L[S,S] \det A[\overline{S},\overline{S}] \pmod 2.$$

Moreover, $\det A[F,F]$ *is odd exactly when* $(V(G),F)$ *is a forest and each of its components has odd order, including isolates.*

Proof. Modulo two, $M = A$, and $\Delta_e \equiv d_{L(G)}(e)$, so $M + \operatorname{diag}(\Delta) = L$. After selecting the indicated parameters the product is

$$\prod_{e\in S} x_e(D_e + \Delta_e x_e) \prod_{e\notin S} D_e^2.$$

In characteristic two the squared factors contain only squared variables. They cannot use an S variable, already present once. Each S variable still needs one odd contribution, so all the $|S|$ linear factors must use precisely the S variables, one each. The coefficient separates into the permanents of the two principal submatrices; modulo two these equal the stated determinants.

Let B be the unsigned vertex-edge incidence matrix modulo two. Then $A = B^T B$. If F contains a cycle its incidence columns are dependent and so is its Gram matrix. For a tree component with p vertices, every $(p-1)$-row incidence minor has determinant one, by leaf elimination. Cauchy–Binet gives its Gram determinant $p \pmod 2$. Multiply over components. Isolates contribute the empty determinant one. This proves the forest characterisation. □

For an odd-order tree the constant coefficient is odd. For an even-order tree with an even-degree vertex there is an edge uv with odd Δ whose deletion gives two odd-order components: take a leaf even-degree vertex of the smallest subtree spanning all even-degree vertices, and an edge leaving that subtree at that vertex. The component on the other side has only original odd degrees, and its original degree sum is $2(|U|-1)+1$, forcing $|U|$ odd. Taking $S = \{uv\}$ in the formula gives an odd coefficient; the constant coefficient is even. Similarly an odd-order unicyclic graph has an odd single-parameter coefficient if a cycle edge has endpoints of different degree parity. These are certificates on already known colourable graph families, not new graph-family claims.

20.2 Offset extensions on odd forests

Corollary 20.4. *If each component of a forest F has odd order, arbitrary real offsets $a(v), b(v)$ admit weights $0,1,2$ such that*

$$a(v) + \sum_{e\ni v} x_e, \qquad b(v) + \sum_{e\ni v} (2 - x_e)$$

both distinguish the ends of every forest edge.

Proof. The product of both colour differences has leading homogeneous part $(-1)^m \prod_e D_e^2$. Its all-two coefficient is odd by Theorem 20.3, independent of offsets. Apply (24) over the reals. □

For physical nonnegative integer offsets with $a(v)+b(v)$ even, this extends constructively to any odd-order connected bipartite graph H. Choose the bipartition labelling p or its complement so that $\sum p \equiv \sum a \pmod 2$; odd order makes the two choices have opposite parity. Solve $\sum_{e\ni v} x_e \equiv p(v) - a(v) \pmod 2$ by a spanning tree with non-tree weights zero. The parity sum condition makes the root equation hold. All weights may be 0 or 1, and the final red and blue degrees both have parity $p(v)$. Every internal edge compares opposite parities. Neither this result nor the corollary automatically keeps outside edges legal when boundary degrees change.

20.3 A fixed forbidden-value application

If each vertex of a simple graph H has real offset $a(v)$ and a fixed forbidden set $Z(v)$ with $|Z(v)| \le d_H(v)$, there is a weighting with $a(v) + \sum x_e \notin Z(v)$ at every vertex. Enlarge each set to size $d_H(v)$ and consider the product of all forbidden linear factors. Its highest homogeneous part is $\prod_v (\sum_{e \ni v} x_e)^{d_H(v)}$, with nonnegative coefficients. Selecting each incident variable once at every vertex shows that its all-two coefficient is at least $\prod_v d_H(v)! > 0$. Interpolation proves the assertion. More exactly that coefficient equals

$$\frac{\prod_v d_H(v)!}{2^{|E(H)|}} \#\{\text{orientations of } {}^2H \text{ with indegree } d_H(v) \text{ at each } v\}.$$

Indeed allocating the two occurrences of each variable to its two endpoint factors gives the multinomial expansion; distinguishing the two copies contributes 2! per edge. This also witnesses positivity by orienting one copy each way. With blue offsets too, combine red forbidden values with the reflected blue values; the union must still have size at most $d_H(v)$. Final neighbour degrees that themselves depend on these edge weights are not fixed forbidden values, so this does not prove the conjecture.

20.4 Limitations of the parity certificate

If all original degrees have the same parity, $L = A$ modulo two. A symmetric zero-diagonal matrix has every odd-order principal determinant zero, by pairing a permutation cycle with its reversal. Thus when m is odd every coefficient of C_G is even. This does not imply zero over the integers: the triangle has $c_G = -6$.

More generally for a connected n-vertex graph the line adjacency matrix has rank $n-1$ at odd n and $n-2$ at even n. Indeed the incidence image is the even-coordinate-sum subspace of dimension $n-1$; the kernel of B^T consists of constants, intersecting that subspace in dimension zero or one respectively. If m exceeds twice this rank and degrees have common parity, either S or its complement is larger than the rank in every coefficient formula. All coefficients vanish modulo two. In particular this happens for every connected d-regular simple graph with $d \ge 4$. Such graphs are known to be colourable [3]; the obstruction is to this parity certificate.

Even integer nonvanishing is a substantial demand. On a regular graph $\Delta = 0$, write $Q = \prod_e D_e$. If C_G is nonzero then Q has a nonzero monomial x^α with all $\alpha_e \le 2$, since otherwise no product with Q can supply the all-two monomial. For arbitrary real vertex offsets the same coefficient survives in $\prod_{uv}(a(u) - a(v) + D_{uv})$. Interpolating on an arbitrary three-element list separately assigned to every edge gives a proper neighbour-sum weighting with those offsets. Thus this nonvanishing would imply the corresponding $(1,3)$ list-weighting property. Ordinary shared-list 1–2–3 weightability does not establish it. We leave the general nonzero-polynomial question open.

21 A chromatic high-minimum-degree condition

We use the following published degree-constrained subgraph theorem of Addario-Berry, Dalal and Reed [5], Theorem 5 in the author version. For a simple host graph with degrees $d(v)$, integers a_v, b_v satisfying

$$0 \le a_v \le \lfloor d(v)/2 \rfloor \le b_v < d(v), \quad b_v \le (d(v) + a_v)/2 + 1, \quad b_v \le 2a_v + 3 \tag{26}$$

admit a spanning subgraph F with $d_F(v) \in \{a_v, a_v + 1, b_v, b_v + 1\}$. We quote this theorem and prove the residue and boundary consequences below.

Theorem 21.1 (D1). *If a simple graph has a proper c-vertex-colouring, $c \geq 2$, and*

$$\delta(G) \geq T(c) := 8c(2c+1) - 7,$$

then it admits doubled weights $x_e \in \{0,1\}$ for which both colour degrees distinguish the ends of every original edge. The same internal conclusion holds with arbitrary prescribed physical external counts at every vertex.

Proof. Put $p = 2c, q = 2c+1, M = pq$, and let $j(v) \in \{0, \ldots, c-1\}$ be the proper colours. With external counts A_v, B_v and internal degree $d = d_G(v)$ choose a residue ϕ_v by

$$\phi_v \equiv 2j(v) - A_v \pmod{p}, \qquad \phi_v \equiv B_v + 2d - 2j(v) \pmod{q}.$$

The moduli are coprime. Set $t = \lfloor d/2 \rfloor$, $r = (t - \phi_v) \bmod M$, $a = t - r$, and $b = a$ if $r = 0$, otherwise $b = a + M$. Both a, b have residue ϕ_v and bracket t. The assumption is $d \geq 4M - 7$, so $\lfloor d/2 \rfloor \geq 2M - 4$ and $\lceil d/2 \rceil \geq 2M - 3$. For $r > 0$,

$$a \geq M - 3, \quad b = a + M \leq 2a + 3, \quad 2b \leq d + a + 2 \iff \lceil d/2 \rceil \geq 2M - r - 2,$$

and the last inequality holds since $r \geq 1$. Also $b \leq t + M - 1 \leq d - 1$. For $r = 0$, all conditions are immediate. Thus (26) holds in the original simple graph.

Give F edges weight 1 and all others zero. For some $\epsilon_v \in \{0,1\}$ the final red and blue residues are

$$R(v) \bmod p \in P_{j(v)} := \{2j(v), 2j(v)+1\}, \qquad B(v) \bmod q \in Q_{j(v)} := \{2j(v), 2j(v)-1\}.$$

The P_j are pairwise disjoint modulo $2c$. The Q_j are also pairwise disjoint modulo $2c+1$: $Q_0 = \{0, 2c\}$, and the others are $\{1,2\}, \{3,4\}, \ldots, \{2c-3, 2c-2\}$. Different proper colours thus give unequal actual degrees in both colours. The choice of external offsets changes no degree inequality, proving the interface version too. □

For $c = 2, 3, 4, 5$ the thresholds are $73, 161, 281, 433$. They are sufficient bounds, with no optimality claim. The theorem has been applied to a simple host only: on two vertices with 100 parallel edges, the parameter pairs $(25, 50)$ and $(30, 55)$ satisfy the numerical inequalities (26), but the allowed factor-degree sets at the two ends are disjoint, so no such multigraph factor exists. Substituting the doubled host into the quoted theorem is therefore not justified by these numerical conditions.

21.1 One internal boundary vertex

Suppose a connected induced region H has all its outside edges at one vertex w. Fix every edge outside the interior of H and assume that outside–outside edges are legal using their final counts. If h boundary edges include m mixed ones, a pure boundary edge forbids at most one residue label and a mixed edge at most two, since the P_j and Q_j are disjoint. Let $C = \max\{2, \chi(H), h + m + 1\}$. At least one of the C labels avoids all outside colour counts. Map the old colour class of w to that label and inject the other proper classes into remaining labels. If $\delta(H) \geq T(C)$, Theorem 21.1 fills H and validates the boundary without changing any given edge.

In particular, a vertex-minimal counterexample cannot have a bridge with one side H of minimum internal degree at least $T(\max\{\chi(H), 3\})$. Replace H by a leaf and use minimality, then the preceding absorption with $h = 1, m \leq 1$. If the smaller graph is K_2, set the bridge mixed directly; its outside endpoint has counts $(1,1)$ and no other outside edge to check. This treats the exceptional smaller graph without assuming it colourable.

21.2 Several CRT regions and low-side ports

Let disjoint induced regions $H_i = G[W_i]$ cover W and put $U = V(G) \setminus W$. Fix all edges other than those internal to a single region, including edges between different regions. Assume all U–U edges are legal with their actual fixed endpoint counts. Given a proper c-colouring j of the *entire* graph $G[W]$, let h_v be the number of edges from $v \in W$ to U, and m_v the number of mixed ones. For every nonempty colour class define

$$L_a = 1 + \max_{j(v)=a} (h_v + m_v), \qquad C = \sum_a L_a,$$

using $c \geq 2$; unused classes may be omitted.

Theorem 21.2 (D2: CRT region absorption)**.** *If every $\delta(H_i) \geq T(C)$, the fixed colouring extends over all regions. If each high vertex has at most t neighbours in U, the uniform sufficient condition is $\delta(H_i) \geq T(c(2t+1))$.*

Proof. Reserve disjoint blocks of L_a labels for the proper colour classes. At v, outside counts prohibit at most $h_v + m_v < L_{j(v)}$ labels. Choose an allowed label in its reserved block, independently at each vertex. All high–high edges, including inter-region edges, still have different labels. Feed all fixed incident contributions as the external counts into Theorem 21.1 in each region. Internal and inter-region edges distinguish both residues; high–low edges avoid their fixed prohibited residues. Low–low edges and all their endpoint counts persist. The estimate $C \leq c(2t+1)$ gives the uniform bound, using additional unused labels if necessary. □

There is a convenient way to supply the low precolouring. For every component D of $G[U]$, attach a private placeholder leaf for each actual high–low edge, at its low endpoint. Different ports use different leaves even if their actual high endpoints agree. A legal colouring of this placeholder graph transfers the placeholder edge weights to the actual boundary edges, preserving each low vertex degree in both colours. If D is bipartite or a cactus, its placeholder graph remains in a class covered by [1, 2]. The only K_2 placeholder case is a single low vertex with one port; make that port mixed, with low counts $(1, 1)$ and no low–low edge. Thus no K_2 colouring is invoked. Inter-region edges can be fixed arbitrarily before absorption. This gives complete infinite families with global minimum degree one or two, rather than only globally high-degree graphs.

21.3 Trees attached to an ordinarily irregular core

Proposition 21.3 (D2: matched tree attachment)**.** *Suppose H is connected, has an edge, and its ordinary degrees differ at the ends of every edge. Let $S \subseteq V(H)$ have a matching to $V(H) \setminus S$ covering S. At each vertex of S attach one disjoint arbitrary rooted tree by a new bridge. The doubled expanded graph has a legal two-colouring.*

Proof. Give matching edges and the new bridges weight 1, all other core edges zero. Core counts are

$$(R, B) = \begin{cases} (2, 2d_H(v)), & v \in S, \\ (1, 2d_H(v) - 1), & v \text{ is a matching helper}, \\ (0, 2d_H(v)), & \text{otherwise}. \end{cases}$$

Matching red edges join degrees two and one. Blue degrees at helpers are odd, elsewhere even; within either parity group ordinary degree irregularity makes them unequal. Each tree bridge meets final core counts $(2, 2d_H(v))$. A singleton tree has counts $(1, 1)$; a larger tree has three mixed-port candidates to avoid the two bridge exclusions by Lemma 11.1. Fill every tree independently. □

A single specified root always has such a matching, by choosing one neighbour. Arbitrary many roots need not, and the resulting expanded graph need not remain ordinarily locally irregular, so unconditional repeated attachment does not follow. Moreover a *given* old colouring can be rigid. Take edges vu, vw, two two-edge u–z paths and three two-edge w–z paths. Its degrees at v, u, w, z are $2, 3, 4, 5$ and the five interiors have degree two. All-mixed edges give a legal old colouring. A leaf at v has no possible new weight preserving that colouring: weights zero, one and two cause respectively a blue equality with w, both equalities with u, and a red equality with w. Changing old vw to zero and giving the new edge weight 1 repairs the graph.

22 Dense graphs with antitone colour degrees

The second global degree condition uses Tutte's factor theorem directly on a doubled graph, where parallel edges are allowed by that theorem. This is separate from the simple-host theorem used in Section 21.

Theorem 22.1 (D3)**.** *Let G be simple with n vertices and*

$$8\delta(G) \geq 7n + 6.$$

Then ${}^2G$ has a two-colouring in which all vertex red degrees are pairwise distinct and all vertex blue degrees are pairwise distinct.

Proof. Put $q = n - 1 - \delta(G)$, so the condition is $n \geq 8q + 14$. Order $v_0, \ldots, v_{n-1}$ by nonincreasing ordinary degree and put $L = \lfloor (n-1)/2 \rfloor$. Set $f(v_i) = L + i$, except that if the sum is odd, increase only $f(v_{n-1})$ by one. The increment occurs exactly when $n \equiv 2 \pmod 4$; only evenness is needed below. Their sum is even and the targets are strictly increasing. Their physical range follows explicitly from

$$0 \leq f(v_i) \leq L + n \leq 2(n - 1 - q) \leq 2d_G(v_i),$$

where $L \leq (n-1)/2$ and $n \geq 8q + 14$ imply the middle bound. If ${}^2G$ has an f-factor, use it for red. Then $B(v_i) = 2d_G(v_i) - f(v_i)$ is strictly decreasing, proving the asserted strong conclusion.

We verify the factor inequalities. For disjoint S, T, put $s = |S|, t = |T|, h = s + t, r = n - h$. Their Tutte left side is

$$D = f(S) + 2\sum_{v \in T} d_{G-S}(v) - f(T).$$

The odd-component term is the number $o_f(S, T)$ of components of $G - S - T$ with odd f sum, since all boundary edges in ${}^2G$ have even multiplicity. Independently of their positions in the order,

$$\begin{aligned} f(S) &\geq sL + s(s-1)/2, \\ f(T) &\leq t(L + n - 1) - t(t-1)/2 + 1, \\ d_{G-S}(v) &\geq n - 1 - q - s. \end{aligned}$$

The last lower bound remains valid when negative. Writing $L = (n-1)/2 - \theta$, $0 \leq \theta \leq 1/2$, gives

$$\begin{aligned} D &\geq (s-t)L + \frac{s(s-1) + t(t-1)}{2} + t(n - 1 - 2q - 2s) - 1 \\ &\geq h\left(\frac{n-6}{4} - 2q\right) - 1 \geq 2h - 1. \end{aligned}$$

For the middle inequality use $s^2 + t^2 - 4st \geq -h^2/2 \geq -nh/2$, $-\theta(s-t) \geq -h/2$, and $t \leq h$.

If $h = 0$, the graph is connected: each component would contain at least $n - q$ vertices and $n > 2q$. Its total f sum is even, so $o_f = 0 = D$. If $h \geq 1$ and $r > 2q$, the remaining graph is connected by the same argument, now with each component of size at least $r - q > r/2$. Thus $o_f \leq 1 \leq 2h - 1$. If $r \leq 2q$, then $o_f \leq r$ and $h = n - r \geq n - 2q \geq 6q + 14 \geq r$, so again $D \geq o_f$. All factor inequalities hold. Tutte's theorem [6] supplies the red factor. □

No optimality of the constant is claimed. The construction may use all three weights $0, 1, 2$, unlike Theorem 21.1. Searching for a particular factor is an implementation task; a search timeout is neither a failed existence proof nor a counterexample.

23 Dense forbidden targets and sequential absorption

Let H have order n and $q = n - 1 - \delta(H)$. At each vertex fix physical external contributions A_v, B_v and a finite set F_v of forbidden *final red degrees*. Put

$$D_v = d_H(v) + (A_v + B_v)/2, \quad \alpha = \min_v A_v, \quad a = \max_v A_v - \alpha.$$

Partition the vertices into equal-D_v classes $C_1, \ldots, C_k$ in strictly decreasing D order and put

$$b_i = \max_{v \in C_i} |F_v|, \quad E = \sum_i b_i + \min_i b_i + 2, \quad W = \max\{a, E\}.$$

Theorem 23.1 (D5: forbidden-target dense interface)**.** *If $n \geq 8q + 4W + 10$, one can colour ^2H so that every final red degree avoids F_v, all final red degrees are pairwise distinct, and all final blue degrees are pairwise distinct. The external contributions are preserved.*

Proof. Let $L = \lfloor (n-1)/2 \rfloor$. Starting at $\alpha + L$, allocate pairwise disjoint consecutive integer blocks in the specified class order. The block for C_i has length $|C_i| + b_i$, except that a class of minimum b_i receives an additional $b_i + 2$ places. Together these blocks form one interval of length $n + E$. In each ordinary class assign distinct targets avoiding the individual forbidden sets by greedy choice; the length exceeds the number of assigned predecessors plus any forbidden set.

In the enlarged minimum class, designate a vertex z. Its interval has at least $2b_i + 3$ places, containing at least $b_i + 1$ integers of each parity. Reserve one allowed even and one allowed odd value for z. Assign the other vertices distinct allowed values while avoiding both reserved places. Writing $n_i = |C_i|$, at the last such step at least $(n_i + 2b_i + 2) - 2 - b_i - (n_i - 2) = b_i + 2$ places remain after all exclusions; if $n_i = 1$ there is no such step. Choose the parity at z that makes $\sum_v (R_v - A_v)$ even. The resulting final red targets are distinct. Order vertices inside each degree class by increasing chosen target. In the resulting global order $v_0, \ldots, v_{n-1}$, D_v is nonincreasing, R_v is strictly increasing, and therefore $2D_v - R_v$ is strictly decreasing.

The n selected places in an interval of length $n + E$ have ordered indices j_i with $i \leq j_i \leq i + E$. For the internal target $f(v_i) = R_{v_i} - A_{v_i}$ we obtain

$$L + i - a \leq f(v_i) \leq L + i + E. \tag{27}$$

The assumption gives $L - a \geq 0$ and $L + n - 1 + E \leq 2(n - 1 - q) \leq 2d_H(v)$, so these are physical internal targets. Their total is even by the reserved parity choice.

For disjoint S, T with s, t, h, r as before,

$$f(S) \geq sL + s(s-1)/2 - as,$$
$$f(T) \leq t(L + n - 1) - t(t-1)/2 + Et.$$

Using $as + Et \le Wh$ and the same degree estimate as in the dense proof yields, without its extra final parity increment,

$$f(S) + 2\sum_{v\in T} d_{H-S}(v) - f(T) \ge h\left(\frac{n-6}{4} - 2q - W\right) \ge h.$$

For $h = 0$ the connected graph has even total target, so its odd term is zero. For $h \ge 1, r > 2q$ the remaining graph is connected and has odd term at most one. For $r \le 2q$ it has at most r odd components, while $h \ge n - 2q \ge 6q + 4W + 10 \ge r$. Thus every factor inequality holds on ${}^2H$. The red factor realises f, and adding the fixed external counts realises the chosen final R and $2D - R$. Their order and forbidden-value choices prove the theorem. □

Theorem 22.1 remains useful: with no external data or forbidden values it needs $n \ge 8q + 14$, whereas the general theorem has $E = 2$ and needs $n \ge 8q + 18$. It is therefore not removed as a supposedly weaker duplicate.

23.1 Preserving arbitrary fixed boundary colours

Let $H = G[W]$ be induced, all outside and boundary weights fixed, and all outside–outside edges legal using their actual counts. For each fixed edge vu, $v \in W, u \notin W$, put in F_v the value $R(u)$ if $x_{vu} > 0$, and the value $2d_G(v) - B(u)$ if $x_{vu} < 2$. Let A_v, B_v be its total fixed incident contributions. A final red degree avoiding F_v makes every boundary comparison legal, while $D_v = d_G(v)$. Theorem 23.1 therefore gives a preserved-colour extension under its stated bound.

If each region vertex has at most t outside edges, then $a \le 2t$, $b_i \le 2t$, and the possible total degrees lie between $n - 1 - q$ and $n - 1 + t$. Thus $k \le q + t + 1$ and $W \le 2t(q + t + 2) + 2$. A uniform sufficient condition is

$$n \ge 8q + 8t(q + t + 2) + 18. \tag{28}$$

The bound uses the per-vertex number of boundary edges, not their total number or the degrees of their outside endpoints. For $q = 0, t = 1$ it is 42, and for $q = 3, t = 1$ it is 90. The actual class-dependent bound can be smaller, for example in the supplied 34-vertex clique witness with fixed mixed ports.

Corollary 23.2 (Sequential dense absorption)**.** *Let $W_1, \dots, W_s$ be disjoint induced regions. Fix every edge not internal to one region, including all inter-region edges. Assume the low–low edges outside the regions are legal with their fixed final counts. If each region satisfies (28) with its internal complement degree q_i and per-vertex outside edge bound t_i, all region interiors can be filled in any order, preserving all fixed edge weights. The sharper actual W_i condition of Theorem 23.1 may be used instead at each step.*

Proof. At a region include all fixed outside incidences in its offsets, including those toward unprocessed regions. Their contributions are known even if the other endpoints' final degrees are not yet known. Forbid conflicts with low vertices and already processed regions, whose degrees are final. Defer comparisons toward future regions. The dense theorem fills the present interior, verifying all its internal and earlier boundary edges. Later fillings cannot alter these degrees or the fixed boundary weights; they enforce the deferred comparisons at their own steps. Induction over the chosen order verifies every edge. The uniform bound allows the worst case of at most $2t_i$ forbidden values per vertex at every step. □

The placeholder-leaf preprocessing from Section 21.2 supplies low-side colours when its components are bipartite or cacti, with the singleton-port exception handled explicitly. Unlike the CRT

region theorem, sequential dense absorption requires no proper colouring of the entire union of high regions. Neither theorem asserts that an arbitrary graph admits a partition meeting its degree hypotheses.

24 Four-vertex complement lifts

Theorem 24.1 (D4: complement-path lift)**.** *Let a simple graph H of order $n \geq 3$ have a given legal doubled colouring and $\Delta(H) \leq n-3$. Let uv be a nonedge whose endpoints have different ordered colour-degree pairs. In the complement replace uv by the five-edge path $uabcdv$ and take the complement again, obtaining H^+. Its colouring can be extended preserving all old edge weights and increasing every old red and blue degree by four. Moreover $\Delta(H^+) \leq |H^+| - 3$, and all five edges of the new complement path have endpoints with different ordered colour-degree pairs.*

Proof. First suppose $B_H(u) \neq B_H(v)$. Give the new edge uv weight zero. For every old vertex w give wb, wc weight 2 and wa, wd weight zero whenever those edges exist; ua and vd are absent. Among new vertices the actual edges are ac, bd, ad; give them weights $1, 2, 0$. Old vertices different from u, v gain $(4, 4)$ from their four new edges. At u, v, the extra old–old edge uv replaces the missing blue contribution at ua, vd, so they also gain $(4, 4)$. All old comparisons persist and uv is legal.

The new counts are

$$a : (1, 2n+1), \quad b : (2n+2, 0), \quad c : (2n+1, 1), \quad d : (2, 2n).$$

Every old final colour count is at most $2(n-3)+4 = 2n-2$. Consequently pure red old–b, c edges and pure blue old–a, d edges are legal. The three actual new–new edges are legal by their displayed counts. The five missing path edges have different ordered pairs as well, by the same strict bounds and the distinct new pairs. Each old degree increases by four, and every new degree is $n+1$, which proves the maximum-degree assertion. If only the red counts of u, v differ, complement the old colours temporarily, apply this construction, and complement the entire result back. Old weights are again preserved and all stated properties remain true. □

Different boundary pairs are a real hypothesis. On C_5 the cyclic word $(0, 0, 1, 1, 2)$ is legal, but two nonadjacent vertices have the same pair $(2, 2)$. A construction adding the same $(4, 4)$ to each and adding an actual edge between them cannot make that edge legal with any weight. This restricts this lift interface, not the graph.

24.1 An explicit nonregular infinite family

For odd $k \geq 3$ let $F_k = C_{2k+1} + 02$, using cyclic labels $0, \ldots, 2k$, and $G_k = \overline{F_k}$. Its two chord endpoints have degree $2k-3$ and the other vertices degree $2k-2$; its size is $2k^2 - k - 2$. For $k = 3$, in the edge order

$$03, 04, 05, 13, 14, 15, 16, 24, 25, 26, 35, 36, 46$$

use weights

$$2, 2, 2, 0, 0, 0, 0, 0, 2, 0, 1, 0, 1.$$

The vertex red degrees are $(6, 0, 2, 3, 3, 5, 1)$ and blue degrees $(0, 8, 4, 5, 5, 3, 7)$, verifying actual legality. The complement edge 60 has different blue endpoint counts. Replacing it by a five-edge path produces G_5 with new labels extending the cyclic order, keeping chord 02. Repeat on the last new edge incident with 0. Its new endpoint blue degree is $2n$, greater than the old endpoint final

bound $2n-2$, so the interface stays available. The lift proves colourability of every G_k, k odd, by induction.

This family also limits a parity-refined defective partition. Suppose one requires a partition (A,B) with

$$\Delta(G_k[A]) \le k-1, \quad \Delta(G_k[B]) \le k-2, \quad R|_A \text{ even}, \quad R|_B \text{ odd}.$$

The degree conditions imply minimum induced complement degrees $|A|-k$ on A and $|B|-k+1$ on B. Only two vertices of F_k have degree three; hence these bounds force $|A| \le k+2$, $|B| \le k+1$, and $|B| \in \{k-1, k, k+1\}$. A minimum-degree-two induced subset of F_k with fewer than $2k$ vertices can only be its triangle 012: if it contains any vertex on the other arc from 2 to 0, its two neighbours are forced, propagating along that entire arc of $2k$ vertices. If $|B| = k-1$, then A has size $k+2 \ge 5$ and induced minimum degree two, impossible; if $|B| = k+1$, the same holds for B of size at least four. Therefore $|B| = k$ is odd, contradicting the even red handshake sum. The degree partition itself exists: $B = \{0, \dots, k-1\}$ and its complement both have no isolates in F_k, which verifies the required complement degree bounds. It is the appended parity prescription that fails. The legal colourings above show that this is not a conjecture counterexample.

24.2 Lifting complements of cubic graphs

Let $F \neq K_4$ be a connected edge-transitive cubic simple graph of order at least six and $H = \overline{F}$. Each component of H is regular of degree $n-4 \ge 2$, so it has a legal doubled colouring by [3]. Some edge of F has distinct endpoint pairs: otherwise connectedness of F would make the pair constant on all vertices, contradicting legality of any actual edge of H. An automorphism carries that complement interface to any selected edge of F. Repeated use of Theorem 24.1 therefore handles replacing that selected edge of F by a path of length $4t+1$ and then complementing, for any $t \ge 0$. At $t > 0$ these graphs are nonregular. Edge-transitivity is used to control the selected initial edge and is not silently omitted.

24.3 Restrictions in complement degree two or three

Suppose G is a vertex-minimal counterexample to the general conjecture and every vertex of $F = \overline{G}$ has degree two or three. If $n \ge 11$, F cannot contain a path $uabcdv$ whose four internal vertices have degree two, whose endpoint degrees differ, and with $uv \notin E(F)$. Delete the four internal vertices and replace that path by uv. The resulting complement H has order $N = n-4 \ge 7$, degree between $N-4$ and $N-3$, and is connected because $2(N-3) > N$. Minimality gives a legal colouring of H. The two endpoint ordinary degrees differ, so their ordered colour pairs differ. The lift reconstructs G, a contradiction. It follows that no path starting at an F-degree-three vertex can continue through six consecutive distinct degree-two vertices: use its first five-edge subpath, whose endpoint degrees are three and two; the sixth vertex ensures the missing closing edge.

There is also no cycle component of F of length at least nine. Shorten that cycle by four. Its length remains at least five, so its complement is connected (and is completely joined to any other complement components). Minimality supplies a colouring of the smaller actual graph. Some edge of the shortened complement cycle has distinct endpoint pairs; otherwise every vertex of that cycle would have the same pair, contradicting an actual edge within its complement. Rotate that cycle component to place the interface at the edge to be lengthened and apply the lift. Finally Theorem 22.1 covers this entire complement-degree class when $n \ge 38$ since $q \le 3$. Any hypothetical counterexample in this class therefore has $n < 38$. No exhaustion of that residual finite range, or reduction of all graphs to this class, is claimed.

25 Cut, parity and modular templates

Every obstruction in this and the next sections concerns an additional restriction on a colouring or on a proposed algorithm. We supply actual legal colourings or cite an already established family result where needed to keep that distinction explicit.

25.1 A maximum-cut template with no exceptions

The template chooses a maximum cut (S, T), requires all edges in $G[S]$ to have weight zero, and requires $R(v) \equiv \not\vdash_T(v)$ at every vertex. For any vertex set U the actual handshake identity is

$$\sum_{v \in U} R(v) = 2 \sum_{e \in E(U)} x_e + \sum_{e \in \delta(U)} x_e. \tag{29}$$

With parity-exception set X, the last sum has parity $|U \cap T| + |U \cap X|$. A single bridge of odd required parity must therefore be mixed. This is a necessary boundary constraint, not a guarantee of an extendible interface.

Proposition 25.1. *For every odd $r \geq 3$, take a triangle abc and attach one copy of $K_{r,r}$ to each triangle vertex by a bridge to a distinguished module root. The resulting graph G_r has minimum degree three and a legal doubled two-colouring, but no maximum cut admits the zero-exception template above. Exactly one exception suffices.*

Proof. The graph has $3+6r$ vertices and $3r^2+6$ edges. A maximum cut has size $3r^2+5$, since each module contributes at most r^2, each bridge at most one, and the triangle at most two; orienting the module bipartitions independently attains all these bounds. Every maximum cut therefore cuts every module edge and every bridge. Each module has r vertices in T, odd. With no exceptions, (29) forces each module bridge to have weight 1. The total red sum is even; the three modules contribute an odd number of T vertices, so precisely one triangle vertex is in T. The other two triangle vertices lie in S, their common edge has weight zero, and their edges to the T vertex must both have weight 1 to give them even red sums after the mixed bridges. They both have counts $(2, 4)$, contradicting their common blue edge.

Here is a full legal colouring. Give ab, bc, ca weights $0, 1, 2$ and all bridges weight 1. Then $a : (3, 3), b : (2, 4), c : (4, 2)$. In each module use rows and columns $0, \ldots, r-1$, with root at row zero. In the a module give the root row all weights 1 and all other entries zero; the root is $(r+1, r+1)$, other rows $(0, 2r)$ and columns $(1, 2r-1)$. For the b module start with root entry $(0, 0)$ of weight 1, row 1 entries in all other columns of weight 1, all remaining entries zero, and complement its internal weights. The root is then $(2r, 2)$; row counts are even and column counts odd. For the c module give the root row zero, every other row weight 1 only in column zero, and all other entries zero. Its root is $(1, 2r+1)$; other rows $(1, 2r-1)$, column zero $(r-1, r+1)$, other columns $(0, 2r)$. Row–column parity verifies every module edge and the displayed root pairs verify the bridges. The triangle edges are checked directly.

Choose the maximum cut with $a, b \in T, c \in S$, module roots on opposite sides to their triangle neighbours. The module patterns have precisely the cut parities, as do a, c. Only b is an exception. Thus the minimum exception count in this template is one. □

The same graph also excludes the different prescription $R(v) \equiv d_G(v) \pmod 2$ at *every* vertex, without mentioning a cut. Each module has odd sum of original degrees, so its bridge would again be mixed. The three triangle vertices would then have distinct odd red degrees, necessarily $1, 3, 5$, because they have equal original degree three. Subtracting the mixed bridge contribution would

require internal triangle red sums $0, 2, 4$, impossible at the edge between the zero and four vertices. This new restriction also fails despite the explicit legal colouring in the proof.

25.2 Small boundary modules and parity transfer

There is a seven-vertex module consisting of triangle abc, two-edge pendant paths at b, c, and a fixed mixed outside edge at a. If its internal cut is maximum, all four path edges cross and two triangle edges cross. Under the zero-exception template, the first edge of a two-edge pendant must be mixed: its leaf and triangle vertex lie on the same side, and the middle parity forces that first weight odd. The module has an odd number of T vertices by its mixed boundary, hence one triangle vertex in T. Each of the two S vertices has one mixed outside contribution, so its edge to the T vertex is mixed too. Their common blue edge again joins counts $(2, 4)$. Thus fixed mixed boundaries need not extend even when total parity is compatible.

Put two-edge pendant paths at all three triangle vertices to obtain a nine-vertex graph O. Every maximum cut and zero-exception template fails by the same argument. For clarity, in edge order $ab, ac, bc, aa_1, a_1a_2, bb_1, b_1b_2, cc_1, c_1c_2$ a legal word is $(1, 2, 0, 1, 0, 0, 0, 1, 1)$. With $T = \{b, c, a_1, b_2, c_2\}$ it has exactly the one exception b_2. This small existence statement is subsumed by the stronger minimum-degree-three family above, but its boundary properties differ.

In O choose root a_2 and give it arbitrary outside total red contribution A. Internal maximum-cut template feasibility with no exceptions requires A odd. If the pendant at a instead has length four and its endpoint is the root, feasibility requires A even. To see both assertions, for a pendant of length $2t$ apply (29) to its $2t$ nontriangle vertices. Its first weight q satisfies $q + A \equiv t \pmod 2$. The wrong boundary parity in either case forces $q = 1$ and one triangle vertex in T, repeating the $(2, 4)$ conflict. These are necessary conditions only; sufficiency of every boundary with that parity is not claimed.

25.3 A six-vertex modulo-three obstruction

Take triangles uab and vcd joined by bridge uv. Even though this cactus is three-colourable, one cannot require every present red and blue comparison to be nonzero modulo three. Write the left weights ua, ub, ab, uv as p, q, t, z and put $h = z - t \pmod 3$. The degree-two edge ab forces $p \neq q$. On ua the restrictions are respectively $q + h \neq 2$, $q + h = 1$, $q + h \neq 0$ when $p = 0, 1, 2$. The ub conditions interchange p, q. Unordered pairs $\{0, 1\}$ and $\{1, 2\}$ immediately contradict these restrictions; only $\{p, q\} = \{0, 2\}$ remains. The right triangle obeys the same rule, so $R(u) \equiv R(v) \equiv 2 + z$ and $B(u) \equiv B(v) \equiv 1 - z$. Every bridge weight then fails. In contrast the actual legal word in order $ua, ub, uv, ab, vc, vd, cd$ is $(1, 2, 2, 0, 2, 0, 1)$, giving counts $u : (5, 1), a : (1, 3), b : (2, 2), v : (4, 2), c : (3, 1), d : (1, 3)$. For instance a legitimate red degree difference of three need not survive modulo three. The modulo-two certificate has its own independent restrictions in Section 20.4.

26 Barriers to nonincreasing local descent

26.1 A closed two-state single-edge plateau

On the wheel with centre 0 and rim 123451, use edge order

$$01, 02, 03, 04, 05, 12, 15, 23, 34, 45.$$

The words $x = (1, 2, 1, 1, 0, 0, 2, 2, 0, 0)$ and $y = (1, 2, 1, 1, 0, 0, 2, 2, 2, 0)$ both have $\Psi = 1$. All single-edge changes, allowing even direct $0 \leftrightarrow 2$ changes, have the following conflict values. Each triple gives the result for new weights $0, 1, 2$.

Edge	From x	From y
01	$(4,1,4)$	$(4,1,4)$
02	$(2,4,1)$	$(2,2,1)$
03	$(2,1,4)$	$(4,1,4)$
04	$(4,1,4)$	$(4,1,2)$
05	$(1,2,2)$	$(1,4,2)$
12	$(1,3,2)$	$(1,5,2)$
15	$(2,5,1)$	$(2,3,1)$
23	$(4,3,1)$	$(2,2,1)$
34	$(1,4,1)$	$(1,4,1)$
45	$(1,2,2)$	$(1,3,4)$

The only nontrivial nonincreasing steps exchange x, y. By induction every such process remains in this two-state set. A simultaneous change of 01 from 1 to zero and 12 from zero to 1 gives $(0,2,1,1,0,1,2,2,0,0)$ with $\Psi = 0$. The complete finite table proves a barrier for single-edge monotonicity, including neutral steps; it does not exclude larger simultaneous repairs.

26.2 Strict positive minima after fixed-target optimisation

On a d-regular graph every assignment realising q has

$$F(q) = 2|\{uv : q(u) = q(v)\}|.$$

Suppose equal-target edges are a perfect matching, every available target $q(v) \pm 1$ occurs on at least three neighbours, and every available $q(v) \pm 2$ on at least two. Then $F(q) = n > 0$ and every realisable target at ℓ_1 distance two has larger F. Changing one coordinate by two creates at least two equalities and removes at most one. Changing two coordinates by one creates at least two equalities to unchanged neighbours at each coordinate, removing at most the two old matching edges. Thus no neutral first step is available, even after completely optimising each fibre.

Here is an explicit infinite realisable family for that criterion. For any even $N \geq 4$, take vertices (i,t) with $0 \leq i \leq 26$ and $t \in \mathbb{Z}_N$, target $q(i,t) = i$. Put a perfect matching within each layer by pairing consecutive even/odd t labels. Between layers start with $a_{ij} = 3$ when $|i-j| = 1$, 2 when $|i-j| = 2$, and zero otherwise. Add each increment in the first table below and its reflection $(26-j, 26-i)$, then make a symmetric.

i	j	increment	i	j	increment
0	1	1	0	2	1
0	4	2	0	11	2
0	24	1	1	3	2
1	20	1	5	14	2
6	13	1	7	9	2
8	10	2			

Its rows sum to 12 and $0 \leq a_{ij} \leq 4$. For $i < j$, add the a_{ij} matchings $(i,t)(j,t+s)$, $s = 0, \ldots, a_{ij}-1$. They are distinct for $N \geq 4$. Together with the within-layer edge the graph is 13-regular. Shifts zero and one between consecutive layers make it connected. Equal targets occur exactly on the within-layer perfect matching and the required neighbour counts hold.

To realise q, for $i < j, i + j \leq 26$ set $b_{ij} = 0$ except

(i,j)	$(3,4)$	$(5,7)$	$(6,7)$	$(8,9)$	$(9,11)$	$(10,11)$	$(10,12)$	$(11,13)$	$(12,14)$
b_{ij}	2	3	4	6	1	6	4	4	2

and define $b_{ij} = 2a_{ij} - b_{26-j,26-i}$ for $i + j > 26$; symmetrise. Within-layer weights at $i = 0, \dots, 13$ are

$$(c_0, \dots, c_{13}) = (0, 1, 0, 1, 2, 2, 0, 0, 2, 2, 0, 0, 2, 1),$$

with $c_{26-i} = 2 - c_i$. Direct row addition verifies $0 \le b_{ij} \le 2a_{ij}$ and $c_i + \sum_{j \ne i} b_{ij} = i$. Give shift-$s$ edges between layers weight $\min\{2, \max\{0, b_{ij} - 2s\}\}$ and within-layer edges weight c_i. Each vertex receives total i, as required. The tables are a fixed integer certificate, independently checked in the artifact. Thus for this connected infinite family of order $27N$, $F(q) = 27N$ and every two-unit neighbour has $F \ge 27N + 2$. These regular graphs have other legal targets by [3]. This bad initial target cannot itself arise by subtracting degrees from a proper 1–2–3 neighbour-sum weighting, since its equal-target matching edges have equal original degrees.

26.3 A proper 1–2–3 initialisation can also be trapped

On the five-vertex path use $x = (1, 0, 0, 1)$. Then $w = x + 1$ has proper neighbouring sums $(2, 3, 2, 3, 2)$, but the doubled target $q = (1, 1, 0, 1, 1)$ has $F(q) = 2$. A path target determines its weights uniquely by leaf recursion. Its alternating bipartition sum is zero, so a two-unit target change must be two opposite-balanced unit changes, corresponding to an alternating contiguous path of weight changes. The two adjacent internal zero weights preclude all such paths except the following eight:

New word	F	New word	F
$(0, 0, 0, 1)$	3	$(2, 0, 0, 1)$	3
$(1, 1, 0, 1)$	3	$(1, 0, 1, 1)$	3
$(1, 0, 0, 0)$	3	$(1, 0, 0, 2)$	3
$(0, 1, 0, 1)$	5	$(1, 0, 1, 0)$	5

No nonincreasing two-unit first step exists, even with an initially proper 1–2–3 weighting. The word $(0, 0, 1, 1)$ is legal and has target distance four from q, attaining the least possible nonincreasing first-step distance. This does not prove that distance four is sufficient for general graphs or specially selected initialisations.

27 Fixed-target and radius obstructions

27.1 Bridge locking, capacity and pure odd defects

For a bridge $e = uv$ whose u side U is bipartite with signs ε, fixed internal target demands give

$$x_e = \varepsilon(u) \sum_{v \in U} \varepsilon(v) q(v).$$

Internal contributions cancel. Thus a bridge may be fixed uniquely; on a forest the entire weighting is unique for a fixed target. Take a triangle uab with path uwl, in edge order ua, ub, uw, ab, wl. The word $(0, 1, 1, 1, 1)$ is realisable and comes from a proper 1–2–3 weighting, but the bridge uw joins equal red degree two, so legality would require $x_{uw} = 0$. Its bipartite side forces $x_{uw} = q(w) - q(l) = 1$. No fixed-target repair exists. The same graph has the legal word $(1, 0, 0, 2, 0)$.

For a purely integer obstruction, take triangles $012, 345$, bridge 03, and leaves $26, 57, 38$. In edge order

$$01, 02, 12, 34, 35, 45, 03, 26, 57, 38$$

use $(0,1,1,0,1,1,1,2,2,0)$. Its targets and blue degrees are

$$q = (2,1,4,2,1,4,2,2,0), \quad B = (4,3,2,6,3,2,0,0,2).$$

The sums $q + d = (5,3,7,6,3,7,3,3,1)$ distinguish all neighbours. Only bridge 03 is forced blue, and there are no forced red edges. After deleting that bridge, the two free components each have target sum nine. At $S = T = \varnothing$ the factor certificate is $g = 0, o = 2$. However giving each triangle the word $(1/2, 3/2, 1/2)$, setting the bridge zero and retaining the leaf weights is a continuous legal realisation of exactly q. Hence continuous feasibility at integer targets need not permit integer rounding. A legal integer repair changes $02, 35$ to 2 and the bridge to zero, changing the targets at 2 and 5 by one. The actual new red degrees are $(2,1,5,2,1,5,2,2,0)$.

The parity lower bound alone can also be strict. On edge set $01, 02, 03, 12, 34$ the word $(1,0,0,0,0)$ has all free-component target sums even, but vertex 1 needs red degree one through its only free neighbour 2 of red target zero. The capacity pair $S = \{2\}, T = \{1\}$ has $g = -1$. Thus its quotient join bound is zero while $F(q) = 1$.

27.2 Failure of exchange convexity

For a triangle the targets uniquely determine weights by $x_{ij} = (q_i + q_j - q_k)/2$. The targets $(1,2,3)$ and $(3,2,1)$ are legal. Starting from the first toward the second with step e_0, the possible second steps are e_0 and $-e_2$. The first gives target pair $(3,2,3), (1,2,1)$, each with $F = 2$; the second gives two copies of $(2,2,2)$, each with $F = 6$. Neither satisfies the fixed-cost exchange inequality with initial total zero. In particular the midpoint of two legal targets can be realisable with positive conflict. The favourable classical exchange properties in Section 18 cannot be assigned to the changing conflict-cost function F.

27.3 An unbounded fixed-parity repair radius

Proposition 27.1. *Let even $c \geq 2$ and $k \geq 1$. Take c disjoint cycles of length $L = 4k + 2$ and two nonadjacent hubs a, b, each joined to every cycle vertex. Put $N = cL$. Give the hubs target zero and every cycle vertex target four, realised by cycle weights 2 and hub weights zero. Fix the maximum cut with both hubs in S and alternate S, T around each cycle, and require final red parity zero on S, one on T. The least maximum coordinate change from those targets is*

$$\rho_{\min} = 2\lceil c/4 \rceil,$$

and this bound is achieved by legal doubled colourings. With E parity exceptions and maximum change at most M, necessarily $2M + E \geq c$.

Proof. The cut has size $2N$ and is maximum. If hubs are on opposite sides, the hub edges contribute N and cycle edges at most N. If hubs share a side and t cycle vertices share it, hub edges contribute $2(N - t)$ and cycle edges at most $2t$. The alternating cut reaches $2N$. Each cycle has $2k + 1$ odd-labelled vertices. Its red boundary sum is therefore a positive odd integer by (29). Hence $R(a) + R(b) \geq c$. Both hub targets have required even parity, forcing the larger to be at least $2\lceil c/4 \rceil$.

Write each cycle $s_0, t_0, \ldots, s_{2k}, t_{2k}, s_0$, starting with all cycle weights 2. Change one hub edge to t_0 to weight 1. For $i = 1, \ldots, k$ change both edges incident with s_{2i} along its two adjacent t vertices to 1. The counts are $t_0 : (5,3)$, other t vertices $(3,5)$, even positive-index s vertices $(2,6)$, and other s vertices $(4,4)$. Assign $2\lfloor c/4 \rfloor$ mixed hub edges to a and the rest to b. Hub red degrees are those two even numbers, with blue degrees at least $2N - c \geq 22$. All cycle and mixed hub comparisons

are parity-separated. The pure blue hub edges to same-side s vertices compare a degree at least 22 to one at most six. The maximum change is exactly the asserted radius.

With E exceptions, at least $c - E$ cycles have no exception and each still needs at least one boundary red copy. The two hub red degrees are each at most M, giving $2M \geq c - E$ independently of any local irregularity requirement. □

This is a minimum-degree-four, 2-connected family with no peripheral trees or degree-two chains. The starting target is feasible but is not a legal colouring. The radius conclusion fixes the stated cut parities. Without fixing them, keep hub weights zero and use the cyclic word $(0,1,2,0,1,2)(0,1,1,0)^{k-1}$ on each cycle. Positions two apart differ, so these words are legal; all changes from the initial target are at most four, with unrestricted parity exceptions. Thus the lower bound is not a distance statement about all legal colourings.

Even an everywhere-one centre window is too narrow on C_5: some red degree in any legal colouring is zero or four. If all red degrees were in $\{1,2,3\}$, the number of degree-two vertices would be odd by the handshake sum, and at most two by independence, hence exactly one. The other four form an alternating 1–3 path. An internal degree-one vertex between two degree-three vertices would need two incident weights each at least one, impossible. The word $(0,0,1,1,2)$ supplies a legal colouring. Translating all degrees by fixed equal external offsets gives the corresponding fixed-interface small-window obstruction.

28 Sharp interface limits and multiple marks

The path-length-six threshold in Theorem 11.2 is sharp for arbitrary fixed physical endpoint data. On a five-edge bare path with end weights 1 and both endpoint counts $(3,3)$, write the word $(1,a,b,c,1)$. Each mixed end edge forces $a = c = 1$. The middle edge then joins identical counts $(1+b, 3-b)$ for any b, and fails. The data are physically realised by adding two mixed leaf edges at each endpoint. That whole ten-vertex tree is colourable by Lemma 11.1; only the fixed interface is impossible. The tree and clique capacity obstructions of Sections 19.4 and 16.1 independently show sharpness of their preserved-colour thresholds.

An arbitrary-root-sum interface also fails for general graphs. On triangle $vabv$ with leaf vz, requiring the internal root red sum zero forces all three incident weights zero. The endpoints of ab then have identical red and blue degrees whatever its weight. Sum six is impossible by complementation. Nonetheless the word $(x_{va}, x_{vb}, x_{ab}, x_{vz}) = (0,2,1,0)$ is legal, with $(R(v), R(a), R(b), R(z)) = (2,1,3,0)$ and blue degrees $(4,3,1,2)$. The full clique interface is stronger than ordinary colourability.

Multiple fixed bridges need not preserve the one-bridge candidate counts. On rooted path $r - u - v$, give r external counts $(2,2)$ from two mixed bridges. If the internal word is (a,b), the red degrees are $(2+a, a+b, b)$ and blue degrees $(4-a, 4-a-b, 2-b)$. At $a = 0$ the edge uv forces $b = 0$, then ru fails in blue; at $a = 2$ it forces $b = 2$, then ru fails in red. At $a = 1$ the first edge excludes $b = 0, 2$, leaving exactly $(1,1)$. Thus the internal root candidate set is the singleton $\{1\}$.

Finally consider a cubic-root safe-core requirement that every root red degree belong to $\mathcal{R} = \{0,1,3,5,6\}$ and every marked mixed port have root red degree in $\{1,5\}$. It cannot accept arbitrary many marks. For a two-vertex core with one mixed mark and one pure port at each vertex, a central edge of weight 0 forces both red sums 1, weight 1 allows neither strong marked sum, and weight 2 forces both red sums 5. No choice is legal.

The failure persists with four independent marked roots: take a double-star core with centres u, v, two leaves at each centre, and one mixed and one pure port at every leaf. A strong marked leaf forces its core edge and its pure port to be both zero (red sum 1) or both 2 (red sum 5).

If the central edge has weight zero, safety forces both centre red sums zero; if it has weight 2, safety forces both centre sums six. Both cases conflict on the central edge. If its weight is 1, each centre with two zero leaf edges has sum 1, conflicting with those leaves, and with two 2-edges has sum 5, also conflicting. Its only possible choice is one of each, giving sum three at both centres, again a central conflict. This is a complete parameter argument; its finite assignment domain has $3^5 \cdot 2^4 = 3888$ members. The weak two-mark lemma in A1 imposes different hypotheses and permits certain unexposed roots to have red sums 2 or 4. These strong-mark obstructions therefore do not contradict the subcubic proof.

29 Signed gaps and paired partitions (E1)

The constructions in this section use actual integer colour degrees. In particular, a nonzero difference of four is useful, although it is invisible modulo four. No assumption about an unknown initial colouring is made. The results of Sections 29–32 are independent of the subcubic proof.

Lemma 29.1 (A signed gap with at most two occurrences)**.** *Let $L_i(y) = \sum_{e \in E_i} a_{ie} y_e$, where $a_{ie} \in \{-1, 1\}$, every E_i contains at least four distinct variables, and every variable belongs to at most two different sets E_i. There is a choice $y_e \in \{-1, 1\}$ for which $|L_i(y)| \geq 2$ for every i. Such a choice can be found in polynomial time.*

Proof. Make an auxiliary vertex for each form. A variable occurring twice is an edge between its two forms, with label $\tau_e = a_{ie}a_{je}$; parallel edges are allowed, but there are no loops. A variable occurring once is a boundary variable at its form.

Choose signs $\sigma_i \in \{-1, 1\}$ and call an auxiliary edge good when $\tau_e = \sigma_i\sigma_j$. Start with any signs and flip a vertex whenever it has more bad than good internal edges. Each flip strictly increases the total number of good edges, so there are at most as many flips as internal edges. At termination, the number of good internal edges at each vertex is at least its number b_i of bad edges. Count boundary variables as good and write g_i for this enlarged good count. Then

$$m_i = g_i - b_i = |E_i| - 2b_i \geq 0.$$

For a boundary variable put $y_e = \sigma_i a_{ie}$; for a good edge put $y_e = \sigma_i a_{ie} = \sigma_j a_{je}$. A bad edge contributes opposite values at its ends after multiplying the two rows by σ_i and σ_j. We may therefore select which end receives $+1$. If o_i bad edges select i as their positive end, then

$$\sigma_i L_i = g_i - b_i + 2o_i = m_i + 2o_i. \tag{30}$$

Call a vertex critical if $m_i \in \{0, 1\}$. Since $|E_i| \geq 4$, every critical vertex has bad-edge degree at least two. In a component of the bad-edge graph with a noncritical vertex, root a spanning tree at such a vertex and select each nonroot vertex as the positive end of its parent edge. If all vertices of a component are critical, its minimum degree is at least two, so it contains a cycle, possibly a pair of parallel edges. Select positive ends cyclically around that cycle and use attached rooted trees to give every remaining vertex a positive parent edge. Set the positive ends of unused edges arbitrarily. Every critical vertex now has $o_i \geq 1$, and every noncritical vertex has $m_i \geq 2$. Equation (30) proves the claim. The local flips and all tree and cycle operations have polynomial bounds. □

Odd-length rows are permitted; their sums have odd parity, so their absolute values are actually at least three. This distinction will remove a boundary obstruction on an even-order path.

Theorem 29.2 (Paired vertex partition)**.** *Let $V(G) = S \sqcup T$ be a supplied partition. Suppose that both induced graphs have maximum degree at most one, that $G[T]$ has no isolated vertex, and that every edge uv internal to either side satisfies*

$$\min(d(u), d(v)) \geq 3, \qquad |d(u) - d(v)| \leq 1.$$

Then $\operatorname{lir}(^2G) \leq 2$*. The mixed original edges can be exactly $E(G[T])$. No degree bound is imposed on an internally isolated vertex of S, and there is no upper bound on the maximum degree of G.*

Proof. Give edges of $G[S]$ weight zero, edges of $G[T]$ weight one, and each crossing edge e weight $1 + y_e$, where $y_e \in \{-1, 1\}$. For each internal pair uv form

$$L_{uv} = \sum_{\substack{e \text{ crossing} \\ e \ni u}} y_e - \sum_{\substack{e \text{ crossing} \\ e \ni v}} y_e.$$

The variables in a row are distinct and number $d(u) + d(v) - 2 \geq 4$. Each crossing edge occurs in at most one row at each end, hence at most twice altogether. Apply Lemma 29.1. The equal internal contributions at u, v give

$$R(u) - R(v) = d(u) - d(v) + L_{uv}, \qquad B(u) - B(v) = d(u) - d(v) - L_{uv}. \tag{31}$$

Both differences are nonzero. On crossing edges, vertices of S have even red and blue degrees, whereas vertices of T have odd red and blue degrees. Thus all crossing edges are legal as well. □

The assumption on degree differences concerns the endpoints of the internal pairs, not the degree range of the whole graph. The artifact includes a connected nonbipartite example with a leaf and degrees ranging from one to six.

29.1 An integer repair with a fixed mixed set

To see why retaining integer values matters, take the cube on $0, \ldots, 7$, with adjacency given by exclusive-or differences $1, 2, 4$, and add 03. Put $S = \{0, 1, 6, 7\}$, $T = \{2, 3, 4, 5\}$ and $M = \{23, 45\}$. The following weights and degrees are a complete witness.

e	01	02	04	13	15	23	26	37	45	46	57	67	03
x_e	0	2	2	0	0	1	2	0	1	2	0	0	0

v	0	1	2	3	4	5	6	7
$R(v)$	4	0	5	1	5	1	4	0
$B(v)$	4	6	1	7	1	5	2	6

With this same M, the stronger demand that both differences on every mixed edge be nonzero modulo four is impossible: on 23 the red degrees are odd, while the ordinary degrees differ by one. A nonzero red difference modulo four would have to be two, making the blue difference zero modulo four. In the displayed legal colouring the actual red difference is four and the blue difference is minus six.

There is also an infinite global obstruction to that stronger demand. For $t \geq 1$, take $H_t = C_{4t}$ with its antipodal matching. Subdivide every edge once, let S be the $4t$ old vertices and T the $6t$ edge vertices, and add a perfect matching inside each side. The resulting graph F_t has $10t$ vertices and $17t$ edges; its S degrees are four and its T degrees are three. It is connected and nonbipartite: an internal S matching edge closes an even path in the subdivision into an odd cycle. Theorem 29.2 colours every F_t with exactly the T matching mixed.

For the same mixed set, the stronger modulo-four condition is feasible if and only if t is even. Indeed, write $R(v) \equiv p_v + 2q_v \pmod 4$, with $p = 0$ on S and $p = 1$ on T. Every internal pair must have opposite q values. The $5t$ pairs force $\sum_v q_v \equiv 5t \pmod 2$. On the other hand, writing every nonmixed weight as $2z_e$ gives $q_v \equiv \sum_{e \ni v,\, e \notin M} z_e \pmod 2$, so the same sum is even. Conversely, when t is even, assign opposite q values on each pair. Their even total can be realised as vertex parities by a spanning-tree leaf elimination in the connected graph $F_t - M$. The resulting weights satisfy the stronger condition. Thus every odd t supplies an integer repair on the same graph and with the same mixed set, rather than a repair obtained by changing the instance or the interface.

30 Path and cycle sides in degree three and four (E2)

Here and in Section 31, every ordinary degree is three or four. An internal degree always means degree in an induced side of the specified partition.

Lemma 30.1 (A useful maximum-cut shape)**.** *Every connected graph with $3 \le d(v) \le 4$ has a partition $S \sqcup T$ whose crossing graph is connected, such that $G[S]$ has maximum degree two, every internally degree-two vertex has ordinary degree four, and $G[T]$ is a matching together with isolated vertices.*

Proof. Among maximum cuts choose one maximising the number of ordinary degree-four vertices in S. A vertex cannot have more than half of its neighbours on its own side. Thus ordinary degree-three vertices have internal degree at most one and degree-four vertices at most two. If a degree-four vertex in T had internal degree two, flipping it would preserve the cut size and improve the secondary objective. Consequently $G[T]$ has maximum degree one. If the crossing graph were disconnected, some original same-side edge would join two of its components, since G is connected. Flipping all vertices of one such component loses no crossing edge and gains all its boundary edges, contrary to maximum cut size. □

This existence argument does not supply a polynomial-time maximum-cut algorithm. The colouring theorems below assume their partitions are given. They do not prove that the isolated vertices permitted by Lemma 30.1 can always be handled.

Theorem 30.2 (Path/cycle side against a matching)**.** *Suppose $d(v) \in \{3, 4\}$ and a supplied partition $S \sqcup T$ satisfies: $G[T]$ is a matching covering T, $\Delta(G[S]) \le 2$, and every internally degree-two vertex of S has ordinary degree four. Then $\operatorname{lir}(^2G) \le 2$, with mixed set exactly $E(G[T])$. The colouring can be constructed in polynomial time.*

We prove the local completion facts first. Give every crossing edge weight $1 + y_e$ with $y_e \in \{-1, 1\}$, every T matching edge weight one, and each S internal edge weight $2z_e$ with $z_e \in \{0, 1\}$. The crossing comparisons are automatically separated by parity. For every T pair impose the row condition $|L_{uv}| \ge 2$ from (31). It suffices to add a matching of selected S edges to these rows, then complete the internal pure edges.

30.1 Binary relations

A relation on two bits has full projections if either variable, when fixed, has some compatible value for the other. Equality, inequality, the exclusion of one ordered pair, and the unrestricted relation all have this property. On a variable path, full projections permit an extension from any one prescribed bit. On a variable cycle consisting only of equality and inequality relations, consistency is equivalent to an even number of inequalities. A variable cycle of full-projection relations is also

consistent if at least one relation allows three or four pairs. To prove the last claim, compose the relations around the cycle. Full projections are preserved under composition; composing a relation with at least three pairs with any full-projection relation still gives at least three pairs. The resulting relation on the first variable contains a diagonal pair. These statements include a two-edge cycle of parallel relations.

30.2 Completing a cycle

Let an S cycle be $v_0 \dots v_{n-1} v_0$ and put $e_i = v_i v_{i+1}$, with indices modulo n. Each vertex has two crossing edges. Let s_i be its number of wholly red crossing edges. Then

$$R(v_i)/2 = s_i + z_{i-1} + z_i, \qquad B(v_i)/2 = 4 - s_i - z_{i-1} - z_i.$$

Whatever the colour of e_i, its legality is exactly

$$s_i - s_{i+1} + z_{i-1} - z_{i+1} \neq 0. \tag{32}$$

When the first difference is zero this is inequality of the two bits; when it is 1 or -1 it excludes one ordered pair; when its absolute value is at least two it imposes no restriction. The variable graph uses step two around the cycle. For odd n it is one odd cycle; for even n it consists of two cycles of length $n/2$, including two-edge cycles when $n = 4$. The binary relation facts give the exact alternatives:

- If n is odd, completion fails exactly when all s_i are equal.
- If $n \equiv 0 \pmod 4$, completion is always possible.
- If $n \equiv 2 \pmod 4$, each of the two variable cycles must contain at least one constraint with $s_i \neq s_{i+1}$.

It is therefore enough to select e_0 on an odd cycle, e_0, e_3 on a cycle of length two modulo four, and no edge otherwise. In the second case $n \geq 6$, the two selected edges are disjoint and control different variable cycles. For a selected edge, the crossing row is $L_i = 2(s_i - s_{i+1})$, with four distinct variables. The gap condition ensures the required unequal values.

30.3 Completing a path

Let an S path have vertices $v_0, \dots, v_{n-1}$ and $m = n - 1$ edges, and write $z_{-1} = z_m = 0$. Then

$$R(v_i)/2 = s_i + z_{i-1} + z_i, \qquad B(v_i)/2 = d(v_i) - s_i - z_{i-1} - z_i.$$

An isolated vertex needs no completion. For a two-vertex path, if the ordinary endpoint degrees differ, at least one of the two pure colours is legal: the two failures would require simultaneously $s_0 - s_1 = 0$ and $s_0 - s_1 = d(v_0) - d(v_1)$. If the degrees are equal, select its only edge and require $s_0 \neq s_1$.

For $n \geq 3$, every nonendpoint has ordinary degree four. Edges between such vertices again give (32), producing two variable paths, one on each parity of edge index. At a left endpoint of ordinary degree four, put $\delta = s_0 - s_1$. Its condition is $\delta - z_1 \neq 0$: it fixes $z_1 = 1$ for $\delta = 0$, fixes $z_1 = 0$ for $\delta = 1$, and otherwise imposes no condition. At a degree-three left endpoint, its own bit z_0 must also be considered:

$$z_0 = 1 \Longrightarrow \delta - z_1 \neq 0, \qquad z_0 = 0 \Longrightarrow \delta - z_1 \neq -1. \tag{33}$$

For $\delta = 0$ this is equality of z_0, z_1; for $\delta = -1$ it excludes $(0,0)$; for $\delta = 1$ it excludes $(1,1)$; otherwise it is unrestricted. In each case it is a full-projection relation joining the two variable paths. The right endpoint has the same description after reversing the indices. Thus:

Endpoint degrees	n	Sufficient treatment
$3,4$	either parity	One variable path, at most one prescribed bit.
$4,4$	odd	Prescribed bits lie on different variable paths.
$4,4$	even	Require left $\delta \notin \{0,1\}$.
$3,3$	odd	A relation cycle with an even number $m-2$ of inequalities.
$3,3$	even	Require left $\delta \neq 0$ to make a relation nonbijective.

In the odd-order $3,3$ case, either a relation is nonbijective, or both endpoint relations are equality and all $m-2$ others are inequality. This includes the parallel endpoint relations when $n = 3$.

Select the first internal edge only when n is even and the endpoint degrees are equal. For endpoints of degree three and $n \geq 4$, the row has four variables and is $L = 2\delta$. For endpoints of degree four, it has five variables and is $L = 2\delta - 1$, since the first vertex has three crossing edges and the second has two. The bound $|L| \geq 2$ in the latter case actually gives $|L| \geq 3$ and excludes $\delta = 0, 1$. The two-vertex case was handled separately. Each path selects at most one internal edge.

Completion of Theorem 30.2. All selected S edges form a matching. Combine their crossing rows with the rows for the T pairs. Each row has at least four distinct variables. Each crossing variable appears at most once on each side, because the selected edges and the T pairs are matchings. Lemma 29.1 supplies all crossing signs simultaneously. The preceding completion arguments then colour each S component. The T comparisons follow from (31), and all crossing edges are separated by parity. Local binary completion is equivalently a two-literal clause system, solvable by strongly connected components, so the whole construction is polynomial. □

30.4 Why internal recolouring is a real extension

Take $2t$ disjoint triangles as S and a perfect matching Q between their vertices, with no pair inside a triangle, connecting the triangles as units. Such a matching exists for every $t \geq 1$: allocate the three ports of each triangle according to a connected cubic loopless multigraph on $2t$ units. For each pair $uv \in Q$ introduce $a, b \in T$ and edges ua, ub, va, vb, ab. This gives a connected nonbipartite graph with $12t$ vertices and $21t$ edges, of degrees four on S and three on T. Theorem 30.2 applies with all ab mixed.

If all S edges were blue while this mixed set stayed fixed, each $K_{2,2}$ crossing block would require a vertex of S with exactly one red crossing edge. Otherwise both S vertices send equal contributions to the two ends of the mixed edge ab. The $3t$ blocks use disjoint S pairs and require at least $3t$ such vertices. In a wholly blue triangle there can be at most one: two would both have blue degree six on their common blue edge. There are only $2t$ triangles, a contradiction. The new theorem restores legality by changing internal pure colours, while preserving the mixed set and the original graph.

The partition requirement itself is not universal. In $K_{3,3}$ plus one chord on one side, an induced matching covering T can contain at most two vertices. Choosing the chord leaves an S vertex of internal degree three; choosing a crossing edge leaves two ordinary degree-three vertices of internal degree two. Taking $T = \varnothing$ also fails. This six-vertex graph has a stored legal colouring; it is an applicability obstruction, not a counterexample to the conjecture.

31 Isolated vertices of the matching side (E3)

Keep $d(v) \in \{3, 4\}$ and the same conditions on $G[S]$, but now allow $G[T]$ to consist of a matching and an internally isolated set I. Let C be the crossing graph. Let F be the selected S matching from the path and cycle proof, and let P be its set of endpoints.

Theorem 31.1 (A protected parity join)**.** *Put $A = C[(S \setminus P) \cup I]$. If every component of A contains an even number of vertices of I, then $\operatorname{lir}(^2G) \leq 2$. The original T matching remains mixed, and the only additional mixed edges belong to A. The component condition is necessary and sufficient for an additional edge set in A with odd degrees on I and even degrees on $S \setminus P$.*

Proof. The last assertion is the elementary parity-subgraph criterion. Necessity follows by summing degrees in each component. For sufficiency, root a spanning tree in each component and determine parent edges from leaves upwards; the root equation follows from the even number of odd targets. Let N be the resulting edge set and make exactly N and the original T matching mixed. Every T colour degree is odd and every S colour degree is even, so crossing edges are automatically legal.

No N edge touches a T pair or a protected S endpoint. Consequently the rows for T pairs and selected S edges keep all their original pure crossing variables, with no new constants. They still have at least four terms and at most two occurrences per variable. Apply Lemma 29.1 and then the preceding component completion. At an unprotected vertex the even number of mixed crossing edges still makes its crossing red contribution even; the local completion only needs the protected row conditions. This proves all required comparisons. □

A component of A with an odd number of I vertices is a certificate against this protected-join restriction. It does not certify that the graph has no locally irregular two-colouring.

For example, choose odd $t \geq 1$ and $r \geq 0$. On $2t$ auxiliary vertices take a connected bridgeless cubic loopless multigraph: three parallel edges for $t = 1$, or C_{2t} with an antipodal matching for $t \geq 3$. Subdivide one edge by $2r$ new vertices. Use these auxiliary vertices as T, replace each auxiliary edge by a new S vertex joined to its ends, and join the $3t + 2r$ vertices of S into one path. Pair the $2r$ new T vertices consecutively by added internal edges. This is a simple graph with $5t + 4r$ vertices and $9t + 7r - 1$ edges. Its S path has odd order, so $F = \varnothing$; the auxiliary graph with the designated edge removed remains connected. Thus the main component of A contains all $2t$ isolated T vertices and the theorem applies. For $r > 0$, internally paired and isolated T vertices coexist with unbounded counts.

31.1 An exact condition with an independent side

Suppose now that T is independent. Fix a crossing mixed set N with

$$d_N(v) \equiv 0 \pmod 2 \ (v \in S), \qquad d_N(v) \equiv 1 \pmod 2 \ (v \in T), \tag{34}$$

and require all other original edges to be pure. Call an S component critical if it is a cycle of length not divisible by four, or an even-order path with both ordinary endpoint degrees three. All its vertices have crossing degree two. Call it saturated when every one of its vertices has N degree two.

Theorem 31.2 (Exact independent-side completion)**.** *For the fixed N satisfying (34), a pure completion of all remaining edges exists if and only if no critical component is saturated.*

Proof. Let c_v be crossing degree and $m_v = d_N(v)$. The crossing red contribution divided by two can be selected independently at each S vertex as any integer in

$$s_v \in [m_v/2, c_v - m_v/2] \cap \mathbb{Z}. \tag{35}$$

Independence holds because every crossing edge has only one S end, and T has no internal comparison. At a nonisolated S vertex the possibilities are $(c, m) = (2, 0), (2, 2), (3, 0), (3, 2)$, giving respectively $s \in \{0, 1, 2\}, \{1\}, \{0, 1, 2, 3\}, \{1, 2\}$.

Saturation of a critical component forces $s_v = 1$ throughout. On a critical cycle, the step-two constraint graph has an odd cycle of inequality relations. On an even-order path with degree-three ends, the two endpoint equalities and the odd number of internal inequalities are inconsistent. This proves necessity, including a two-vertex path.

For each unsaturated critical component choose one vertex with $m_v = 0$, set its s to zero and all other s values to one. On an odd cycle this creates a nonzero neighbouring difference. On a cycle of length two modulo four the two differences around the special vertex occur in different variable cycles. On a path with degree-three ends the change makes an endpoint or internal relation nonbijective (and directly separates the two vertices when the path has order two). The earlier relation argument gives a completion.

For a cycle of length zero modulo four, an odd-order path, or a path with different endpoint degrees, take all $s_v = 1$. An isolated vertex can take any allowed value. The only remaining case is an even-order path with both endpoint degrees four. Its endpoints can each take values one and two. For order two choose these different values. For order $n \geq 4$, set the left endpoint and all internal values to one. Set the right endpoint to one if $(n - 4)/2$ is even, and to two otherwise. The two unary constraints on one variable path are separated by exactly $(n - 4)/2$ inequalities, so this choice is consistent; the other variable path has no conflicting unary constraints. Realise the chosen crossing degrees using (35), then complete the internal components. All crossing edges satisfy parity separation. □

31.2 A bridge obstruction and its repair

The opposite parity configuration, developed more generally in Section 32, gives the following useful special case.

Corollary 31.3 (Even components with an independent opposite side)**.** *If T is independent and every S component is an even-order path or an even cycle, under the degree-three/four assumptions above,* $\mathrm{lir}(^2G) \leq 2$. *There is no parity restriction on $|T|$.*

Proof. Make an internal perfect matching of every S component mixed, and all other internal edges blue. Bipartition each component, and give its two classes respectively zero and two wholly red crossing edges at each vertex. There are at least two crossing edges at every vertex. Red degrees are one and five; every internal red difference has absolute value four, and its blue difference is nonzero because ordinary degrees differ by at most one. All crossing edges are pure, joining odd colour degrees in S to even colour degrees in T. □

An infinite family shows why parity solvability alone does not guarantee the first configuration. For $t \geq 1$, subdivide one edge of C_{4t} plus its antipodal matching, obtaining a graph Q_t of order $4t + 1$ with one degree-two vertex. Join three disjoint copies of Q_t to a new centre through these vertices. The resulting cubic graph H_t has three bridges. Let $T = V(H_t)$ and introduce an S vertex on each edge of H_t, giving the subdivision as the crossing graph. Denote the three bridge vertices in S by w_1, w_2, w_3. Add w_1w_2, make six nonbridge S vertices into a six-cycle, and pair all remaining

S vertices. There is an even number remaining, and there are enough nonbridge vertices for every $t \geq 1$.

This simple connected nonbipartite graph has $30t + 10$ vertices and $45t + 18$ edges; exactly six vertices have degree four, and all others degree three. Every S component has even order, so Corollary 31.3 colours it. But if all S internal edges are pure, S has even colour degrees, and T odd colour degrees, each half of a subdivided bridge is forced mixed. Indeed, its cut side contains $4t+1$ odd targets in T, so summing the mixed degrees modulo two forces the sole allowed crossing boundary edge into N. Thus the crossing red contribution is two at both w_1, w_2. If their common internal edge has weight $x \in \{0, 2\}$, then

$$R(w_1) = R(w_2) = 2 + x, \qquad B(w_1) = B(w_2) = 4 - x.$$

Its actual colour is illegal. The obstruction excludes that entire parity configuration, not just a particular choice of N, while the opposite configuration colours the same infinite family.

32 Odd-component root ports at arbitrary degrees (E4)

The following theorem removes the global degree restriction from Corollary 31.3 and permits odd paths, odd cycles and internally isolated vertices. The needed port certificate is explicit.

Theorem 32.1 (Root-port construction)**.** *Let a finite simple graph have a supplied partition $S \sqcup T$ with T independent and $\Delta(G[S]) \leq 2$. Suppose every nonisolated vertex of $G[S]$ has at least two neighbours in T, and every S internal edge uv satisfies $|d(u) - d(v)| \leq 1$.*

Select one root in each odd-order S component and no root in an even-order component. In an odd path, number vertices from an end and select a root with even index; in a singleton, select its only vertex. In an odd cycle, choose a root whose two neighbours are not both of degree $d(r)+1$ and not both of degree $d(r) - 1$. Let $\mathcal{U}$ be the selected roots. Suppose a crossing edge set N exists with:

- *every edge of N joins a root to T;*
- *every T vertex has even N degree;*
- *every internally nonisolated root has N degree one or three;*
- *every internally isolated root has positive odd N degree.*

Then $\operatorname{lir}(^2G) \leq 2$, constructively. There is no bound on the maximum degree of G or on degrees in T; internally isolated vertices of S may include original leaves.

Proof. An eligible root exists on every odd cycle. Otherwise each vertex would be a strict local minimum or maximum of the degree sequence, with these two types alternating, which cannot close around an odd cycle. Deleting the chosen root from an odd cycle leaves an even-order path; deleting an even-index vertex from an odd path leaves two even-order paths. Give each remaining path or even cycle an internal perfect matching. Make all its matching edges and all edges of N mixed; make every other S internal edge blue.

Each nonroot S vertex has one internal mixed edge and no mixed crossing edge. Every root has an odd number of mixed crossing edges and no internal mixed edge. Thus all S colour degrees are odd, all T colour degrees are even, and all crossing edges are automatically legal.

On each remaining even path or cycle, assign red degrees one and five to the two bipartition classes. A nonroot has at least two pure crossing edges, so these targets are realised by making

zero or two of them wholly red. Internal edges between nonroots have red difference of absolute value four; their blue differences are nonzero because ordinary degree differences are at most one.

Give each nonisolated root red degree three. If it has one port, use one other crossing edge wholly red; if it has three ports, use no other red crossing edge. The assumed two crossing neighbours ensure capacity in the first case. For a root neighbour u, the internal edge ru is blue, and its blue-degree difference is respectively

$$B(r)-B(u)=2(d(r)-d(u))-2 \quad \text{if } R(u)=1, \qquad B(r)-B(u)=2(d(r)-d(u))+2 \quad \text{if } R(u)=5. \tag{36}$$

If $d(r)-d(u)=1$, label five is forced; if it is -1, label one is forced; if it is zero, either works. On an odd path, the components on the two sides of the root can be phased independently. On an odd cycle, the two endpoints after root deletion must receive opposite labels. The only inconsistent requests are that both must be one or both must be five, exactly the two cases excluded by root eligibility. Hence a compatible phase exists.

An internally isolated root needs no internal comparison: give all its remaining crossing edges blue. Its colour degrees retain odd parity, with no need for red degree three. Every crossing edge has only one S end, so all these choices are independent. The preceding arguments cover every original edge and produce the colouring by finitely many polynomial-time traversals. □

Corollary 32.2 (An exact root parity test within capacity)**.** *Fix eligible roots $\mathcal{U}$, and let $A = C[\mathcal{U} \cup T]$ in the crossing graph. A subgraph with odd degrees on $\mathcal{U}$ and even degrees on T exists if and only if each component of A contains an even number of roots. If every internally nonisolated root has crossing degree at most four, this condition guarantees the port set of Theorem 32.1.*

Proof. Apply degree summation and spanning-tree parity recovery as in Theorem 31.1. An odd port count at a nonisolated root of crossing degree at most four is necessarily one or three. Internally isolated roots permit any positive odd count. □

For roots of larger crossing degree the parity condition is still exact for odd/even edge selection, but it alone does not guarantee the one-or-three capacity for nonisolated roots. A failed tree recovery with five ports must not be called infeasibility. A supplied qualified port set remains sufficient at arbitrary degrees.

Pairing roots with common neighbours in T is a special case: give each pair its two crossing ports, making all T counts even. It is not necessary. For odd $b \geq 3$, take a vertex $s_0 \in S$ and b groups of three S vertices; in each group add a private triple of T vertices, join the two triples completely, and join s_0 to one chosen private T vertex. All S components are singletons. The root adjacency graph formed by common T neighbours is b copies of K_4 sharing s_0. It has no perfect matching: deleting s_0 leaves b odd components, which cannot all be matched through a single vertex. Nevertheless the root-port graph is connected and has $1+3b$ roots, an even number, so Corollary 32.2 applies. Explicitly, in each group select the edge from s_0, one other edge to that same private T vertex, and two edges from the other two roots to a common private T vertex. This also shows why internally isolated roots need arbitrary odd port capacity. These particular examples are bipartite, whose ordinary existence theorem is already known [1]; their purpose is to separate two port criteria, not to claim a new bipartite result.

Finally, fixing the roots can still lose a legal colouring. Take $S=\{0,1,2,3\}$, $T=\{4,5,6\}$ with ordered edges

$$04, 15, 24, 26, 35, 36, 23.$$

The odd components give roots $0, 1$; the permitted root-port graph has two components each containing one root, so the root-only scheme fails. Yet the weights $(1, 1, 1, 2, 1, 0, 0)$ give

$$R = (1, 1, 3, 1, 2, 2, 2), \qquad B = (1, 1, 3, 5, 2, 2, 2),$$

which are legal edge by edge. Here the even component $2-3$ also receives crossing mixed ports. Thus the exact root parity test does not solve general even-component transport or produce the required partition for every graph.

33 Evidence and limits of the integer partition results

The four new groups are not four independent solutions of the mother conjecture. E1 contains a general signed tool and an unbounded-degree paired-partition application. E2 enlarges the S side to paths and cycles within degrees three and four. E3 addresses some, but not all, isolated vertices on the matching side and gives an exact fixed-interface criterion when that side is independent. E4 handles odd components under an arbitrary-degree root-port certificate; the even-component existence corollary is retained as its simpler special case rather than counted as an additional maximal theorem. Fixed mixed sets, port counts and parity patterns are part of the statements, not interchangeable notions of extension.

The implementation checkers test 647 signed systems, 15,757 local path/cycle profiles against full binary enumeration, 52 independent-side fixed-join domains, and 727 eligible fixed-root atlas domains. They also check parameter families, the actual integer repairs and the explicitly failed restricted strategies. Separate standard-library auditors, without importing the constructors or solvers, rebuild the saved signed sums, local truth tables, cuts and actual copy degrees. Across the four auditors, the stored whole-graph witnesses account for 972 actual copies: 100, 190, 258 and 424 in the respective archived source audits. The release report supplies machine-readable per-audit counts so that repeated or overlapping examples are not counted as distinct new graph families. These are implementation and certificate audits; the displayed proofs justify the unbounded parameter ranges.

The finite counterexamples to weakened signed-gap hypotheses are also retained. With only three terms, antisymmetric incidence forms of K_4 would require every vertex to be a source or a sink to reach absolute value at least two, forcing a bipartition. Allowing three occurrences already fails for

$$L_1 = y_0 + y_1 + y_2 + y_3 + y_4, \quad L_2 = -y_0 - y_1 + y_2 + y_3 + y_4, \quad L_3 = y_0 + y_1 - y_2 - y_3 + y_4.$$

After a global sign change $L_1 \geq 3$, at most one variable is negative. With none negative $L_2 = 1$; if the sole negative is in positions zero or one then $L_3 = -1$, otherwise $L_2 = -1$. Arbitrary affine shifts fail too: for antisymmetric incidence forms of K_5, $L_v + 1 = 2d^+(v) - 3$. The condition $|L_v + 1| \geq 2$ allows only outdegrees $0, 3, 4$; at most one vertex has outdegree zero, so their sum is at least twelve rather than the ten edges of K_5. These are obstructions to proposed stronger tools, not to locally irregular two-colourability of the doubled graphs.

No claim of exhaustive historical priority is made for these tools. The known regular and bipartite cases remain attributed to their original sources. General graphs still need a suitable partition, compatible mixed-edge choices, and control of the remaining component configurations. The unrestricted conjecture is not proved.

34 Artifacts, verification status and remaining problem

34.1 Three distinct roles of computation

The artifact separates complete proof kernels, explicit graph witnesses, and bounded implementation tests. The first role is logically part of the computer-assisted proofs. A1 uses the complete safe-core relation tables, 163 sparse-core representatives, and a 4,256-case weak two-mark kernel, with fixed-point and path-saturation arguments for unbounded length. The clique interface uses 836 complete small-order offset bases and its two-vertex induction. The decorated-path interface uses all 93 endpoint comparison types, all 96 transition rows, and the displayed lower-bound and saturation identities. These domains are finite because their equivalence reductions are proved, not because experiments stopped at a chosen graph order.

The second role is independently checking submitted weight assignments. The release verifier rebuilds red and blue degrees from the actual edge-copy counts, including physical offsets when present, verifies all required comparisons, checks any recorded final degrees, and confirms preservation of supplied boundary weights. Large stored region witnesses can therefore be audited without repeating their discovery search. The 13-regular descent barrier is supported by a fixed integer table whose row sums and neighbour counts are checked directly; its arbitrary fibre sizes are justified by the explicit matching construction.

The third role is testing the constructive implementations on graphs, offsets and lengths beyond the finite kernels. These checks include small graph atlases, random compressed graphs, long paths and return paths, prescribed-root cliques, and several complete dense-region examples. They audit code, rather than replace any infinite proof. Original research-run reports are retained separately from reports of the new release run. The provenance ledger identifies frozen original notes and the script dependencies. Optional factor searches use NetworkX and sometimes Z3. Their timeouts are reported as unfinished searches; a timeout is never treated as an infeasibility certificate.

34.2 Reproducing the release

The default complete-certificate command, from an extracted archive, is

```
python -I -B verify.py
```

It uses the Python standard library. It invokes the A1 certificate verifier and its independent local recalculation, then the additional clique and decorated certificates, fixed integer tables, stored graph witnesses and independent arithmetic checks. It also runs all four portable E1–E4 audits without importing their constructors or solvers. Those audits recount 972 actual copies in the saved whole-graph witnesses and rederive the submitted rows, local relations, parity cuts and strategy obstructions. Do not run Python with optimisation enabled: the certificate checkers deliberately use assertions. A JSON report can be written using the option `--output validation/certificates.json`.

For constructive implementation tests, install the pinned dependencies in `requirements.txt` and run

```
python -B run_checks.py
```

This runs the released construction checkers and saves a bounded report. The A1 command-line constructor remains available as `construct.py`, with examples and an independent copy-degree checker in the root directory. Other constructors and their scope checks are in `artifact/`; the coverage ledger points to each public implementation. A method-not-applicable response is not a counterexample to the mathematical conjecture.

The editable English source is `manuscript/main.tex` with included section files and an explicit bibliography. It can be compiled by a standard LaTeX installation or Tectonic. No user-specific absolute path is required by the released proof data or verifier. The manifest records SHA-256 digests of the released files. The final validation report records fresh extraction, certificate execution, constructor checks, PDF rebuild and source-reference diagnostics. Checks of the archive are local; no repository deposit, Zenodo upload, DOI allocation, or journal submission is implied.

34.3 Use of AI and research status

This manuscript and its research artifacts were developed with extensive assistance from OpenAI Codex, including mathematical exploration, proof drafting, programming, calculations, internal review and English editing. Automated checks and internal reviews are documented as such; they are not independent expert peer review. No Lean or other proof-assistant verification is claimed.

The artifact attributes the pre-existing graph-family results and the classical degree-list, factor, matching and interpolation tools used in the proofs. Its grouping of results records this research programme's contributions and boundaries; it is not an assertion that every grouped lemma first appears here. A literature-screening record is supplied as provenance, with no guarantee of exhaustive historical priority. The manuscript, documentation and finite certificate data are licensed under Creative Commons Attribution 4.0 International (CC BY 4.0). The accompanying Python source code is licensed under the MIT licence. The licence-scope file and full licence texts accompany the release.

The remaining mathematical problem is the original arbitrary-graph conjecture. None of the positive results forces a general graph to have a sparse, homogeneous, clique, parity-separated, high-degree or dense region structure. Target optimisation is exact only after a target is fixed, and the algebraic route lacks a general nonvanishing theorem. The E1–E4 partition conditions and mixed-port certificates have not been proved to exist for every graph. In particular, a parity-subgraph solution can saturate a forbidden local component, and root-only joins can exclude valid colourings using even components. Minimal-counterexample restrictions still permit arbitrary large, nonregular, minimum-degree-three cores. The explicit obstructions show why several universal parity and monotone-repair assertions cannot fill that gap. A1 covers the full subcubic range; the later conditions extend beyond that range but do not combine into a proof for every graph.

A Finite domains and certificate specification

The proof certificates describe exact relations, not probabilistic tests. The five data files are listed in Table 2. The marked-theta table supports a more general internal helper; Corollary 5.2 makes that helper unnecessary for the single-mark step of Theorem 2.1. It is included so that the implementation's dependencies remain self-contained.

Data file in `artifact/`	Contents
`safe_cactus_interfaces.json`	Six message types, 43 node cases, letter matrices and complete even product layers
`safe_all_cactus_interfaces.json`	The same data, odd product layers and even-cycle rooted profiles
`sparse_bip_core_interfaces.json`	Nine states, six-step walks, 163 sparse-core representative witnesses
`weak_two_marked_interfaces.json`	Arm layers, 4,256 two-mark witnesses and the embedded chain certificate
`marked_theta_interfaces.json`	608 marked-theta witnesses and the chain certificate for the optional helper

Table 2: Finite certificate data. Equality of entire domains and relation sets is checked, rather than equality of recorded totals alone.

A.1 Sparse-core representatives

Put $I = \{1, \dots, 8\}$ and $J = \{4, 6, 8\}$. Length parameters in Table 3 record only chains between forks or around cycles; leaf-arm lengths are handled by the separate induction in Lemma 4.1. A fork is a vertex of core degree three. An even cycle represented by two fork-to-fork paths has lengths a, b.

Shape	Complete parameter domain	Cases
Three-arm tree	No chain parameter	1
Cycle with an arm	Cycle length $a \in J$	3
Double-star tree	Fork-to-fork length $b \in I$	8
Cycle attached to a fork	Cycle length $a \in J$, joining length $b \in I$	24
Cycle with two arms	$a, b \in I$, $a \le b$, $a + b$ even, $a + b \ge 4$	19
Theta	$a \le b \le c$ in I, all of the same parity, at most one equals 1	36
Two cycles joined by a chain	Cycle lengths $a, b \in J$, joining length $c \in I$	72
Total		163

Table 3: Complete finite domain for the sparse bipartite core lemma.

Each record supplies one or two fork triples in $\{0, 1, 2\}^3$, with their sums in $\{1, 3, 5\}$. For every compressed chain the checker starts at its given end state, applies (9) once per internal vertex and checks its other end state and final unequal-degree comparison. Each leaf arm is checked against the relation of possible pure-port leaf completions. The shape data specify how triples are shared by chains, so consistency of all three incidences is explicit.

A.2 The two-mark kernel

Label the direct unmarked position `u0`. Label arms of lengths 1,2 and at least 3 by `u1`, `u2`, `u3`. Use `m0` for a direct marked position and `m1`,...,`m6` for marked arm lengths 1 through 5 and at least 6. Write U and M for these four- and seven-element sets of labels. The complete domain is

$$\Big(\mathrm{Sym}^2(M) \times \mathrm{Sym}^2(U) \;\cup\; (M \times U)^2\Big) \times \{1, 3, 5, 7\}. \tag{37}$$

The two pieces are kept as distinct canonical patterns. Each record contains the two side labels, the middle length, the two end states (x, r) and the actual choices for the two positions at each side.

For a direct position, its contribution is 0 or 2 if unmarked and 1 if marked. A direct marked position requires its root red degree not to equal 3. For an arm position, its pair (y, s) must belong to the message set for that label and satisfy $s \neq r$ at the distinguished root. At either side, the root degree must equal the middle end weight plus its two position contributions. The middle path must satisfy the recurrence (9), allowing actual end red degrees 2 and 4. These are all the checks imposed by `weak_two_marked_certificate.py`.

The independent local verifier does not read these 4,256 witnesses. It reconstructs the arm messages, forms every possible side state, computes all middle-path relations and checks that each member of (37) has a witness. This provides a separate calculation of the finite existence claim, while the main checker validates the portable witness data.

A.3 Execution and trust boundary

From a fresh extraction of the archive run

```
python-Bverify.py
```

The program rejects optimized Python execution, since the supplied checkers use assertions. It verifies the six-state closure, the 163-case sparse-core certificate, the 4,256-case two-mark certificate, the optional 608-case marked-theta certificate and the fresh local calculation. It also rejects deliberately damaged copies of selected certificate data in memory. No source data is changed.

For a full graph example, install the optional dependency and run

```
python-mpipinstall-rrequirements.txt
python-Bconstruct.pyexamples/triangle.json
```

The input gives an integer vertex list and unordered pairs of distinct vertices. Duplicate edges, disconnected inputs, degree above three and K_2 are rejected. The output lists red and blue multiplicities and actual vertex degrees, which are recounted before output. The optional argument `--output` saves the resulting JSON to a chosen path.

All calculations rely on ordinary exact Python execution. They are reproducible computation, not proof-assistant kernel checking. Conversely, the claim of arbitrary size rests on the explicitly proved classifications, finite-state inductions and global compatibility arguments in the body, which cannot be replaced by running a finite suite of graph examples.

B Additional finite certificates

B.1 Decorated path tables

Order the 24 states by degree $D = 2$ then 3, then by outgoing weight $p = 0, 1, 2$ and increasing red degree. Encode a state subset by the sum of powers 2^i at its indices, starting at zero. The complete lower-bound families used in Theorem 11.2 are as follows:

F_1: 9, 16, 36, 65, 72, 130, 260, 288, 52736, 473088,
1132032, 1149440, 1166336, 1688064, 2100224, 2671104,
3217408, 3228160, 3768832, 3784704, 5313536, 6293504,
6754304, 8599552, 8722432, 8792064, 8859648, 10686464,
12914688, 15073280.

F_2: 27, 45, 54, 130, 216, 325, 360, 432, 3264000, 5313536, 7425536, 7588352, 10344960, 10984448, 11463168, 12022272, 15014912, 15126016, 15144960.

F_3: 219, 325, 438, 7458304, 11463168, 15126016, 15145984.

F_4: 365, 12581376, 16244224.

The three final masks are the signatures in the table below; each cell records possible red degrees at the given outgoing weight.

Signature	D	$p=0$	$p=1$	$p=2$
M_A	2	$\{0,2\}$	$\{1,3\}$	$\{2,4\}$
M_L	3	$\{0,2,3,4\}$	$\{1,2,3,4,5\}$	$\{2,3,4,6\}$
M_S	3	$\{0,1,2,3\}$	$\{1,2,3,4,5\}$	$\{3,4,5,6\}$

The local full-state signatures are

$$\begin{aligned} \Omega_A &= (\{0,1,2\},\{1,2,3\},\{2,3,4\}), && D=2,\\ \Omega_L = \Omega_F &= (\{0,1,2,3,4\},\{1,2,3,4,5\},\{2,3,4,5,6\}), && D=3,\\ \Omega_S &= (\{0,1,2,3\},\{1,2,3,4,5\},\{3,4,5,6\}), && D=3. \end{aligned}$$

For every $M \in \{M_A, M_L, M_S\}$ and every letter X, $T_X(M) = \Omega_X$. Every Ω_X contains one of the final three masks. The machine-readable table contains all 372 initial cases, 96 transition rows, 224 successive-family containment checks and 12 saturation checks, with the exact state and endpoint domains. The independent checker reconstructs the local relations rather than trusting the recorded counts.

B.2 Clique bases and additional witnesses

The root clique table is indexed by (n, A, B, S) for $n = 4, 5$, the clipped physical domain in Theorem 16.1, root label zero and lexicographic edge order. Every entry records a full word in $\{0,1,2\}^{\binom{n}{2}}$ and the truth of the nonroot interior bound. The independent checker verifies each bound itself and permits weakness only at the two specified K_4 inputs. The domain has 836 entries and 14,424 actual coloured copies.

The portable datasets include the regular-layer integer table, the decorated-path transfer table, and the complete clique root bases. Their file names and schemas are identified in `COVERAGE.md`. The witness files for dense, offset and sequential regions instead record complete actual graphs, weights, final colour degrees and, where applicable, the fixed boundary edges and offsets. Their verification checks a concrete construction and does not replace the factor-theorem arguments proving arbitrary-size existence.

The standard-library release verifier additionally recalculates the wheel's entire single-edge neighbourhood, the five-vertex path's entire two-unit target neighbourhood, the strong-mark domain, and the nine-vertex continuous/integer distinction using rational arithmetic. It checks the frozen regular-layer table against the explicit formula, and tests coefficient/permanent and modular identities on small exact inputs. Its report records actual domain sizes computed in that run. These are finite identities or witness audits, with their roles kept separate from the unbounded proofs.

B.3 Coverage and dependency files

`COVERAGE.md` maps every retained inventory item and completed post-17:00 note to its English proof, implementation and certificates. `PROOF_DEPENDENCIES.md` lists the exact external statements used and the difference between complete proof kernels and sampled graph tests. `provenance/source_manifest.json` hashes the original top-level source snapshot, and `provenance/source_scope.json` records the scientific scope and snapshot date without internal chat identifiers. Chinese source notes are included for provenance; the English proofs do not require reading them. No unfinished current-thread experiment is promoted to a theorem.

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